

CSES Introductory Problems — Problem Reference

All listed tasks: 1 s, 512 MB. Examples omitted for compactness.

1072 — Two Knights

For every $k = 1, 2, \dots, n$, count ways to place two knights on a $k \times k$ chessboard so that they do not attack each other.

In: n

Out: n integers: the answer for each $k = 1, \dots, n$

Constraints: $1 \leq n \leq 10^4$.

1618 — Trailing Zeros

Given n , find the number of trailing zeros in $n!$. **In:** n

Out: number of trailing zeros in $n!$

Constraints: $1 \leq n \leq 10^9$.

1622 — Creating Strings

Given a lowercase string, generate every distinct permutation of its characters. **In:** one lowercase string s

Out: first the number k of distinct strings; then all k strings in lexicographic order

Constraints: $1 \leq |s| \leq 8$.

1623 — Apple Division

There are n apples with weights p_i . Partition all apples into two groups, minimizing the absolute difference between the groups' total weights. **In:** n ; then $p_1 \dots p_n$

Out: minimum possible weight difference

Constraints: $1 \leq n \leq 20$, $1 \leq p_i \leq 10^9$. Use 64-bit.

1624 — Chessboard and Queens

Place eight queens on an 8×8 board so no two attack each other. Queens may be placed only on free squares; reserved squares do *not* block attacks. **In:** 8 rows of 8 characters: $\cdot$ free, $*$ reserved

Out: number of valid placements

Constraints: Fixed 8×8 board and exactly 8 queens.

1625 — Grid Paths

A path on a 7×7 grid starts at the upper-left square and ends at the lower-left square, visiting each square at most once. Its 48 moves are described by D, U, L, R, or ? (any direction). Count paths matching the description. **In:** one 48-character string over ?DULR

Out: number of matching paths

Constraints: Grid is fixed at 7×7 ; description length is 48.

1743 — String Reorder

Rearrange the characters of an uppercase string so no two adjacent characters are equal. Among all valid rearrangements, find the lexicographically smallest. **In:** one string s over A–Z

Out: lexicographically smallest valid rearrangement, or -1 if impossible

Constraints: $1 \leq |s| \leq 10^6$.

2165 — Tower of Hanoi

Move n disks from stack 1 (left) to stack 3 (right), using stack 2. Move only the top disk of a stack; never place a larger disk on a smaller one. Minimize the number of moves. **In:** n

Out: minimum move count k , then k lines $a \ b$ describing each move

Constraints: $1 \leq n \leq 16$.

2205 — Gray Code

Construct a Gray code of length n : all 2^n bit strings of length n , ordered so consecutive strings differ in exactly one bit. **In:** n

Out: 2^n lines forming any valid Gray code

Constraints: $1 \leq n \leq 16$.

2431 — Digit Queries

Consider the infinite string 123456789101112.... For each query, find the digit at 1-indexed position k . **In:** q ; then q values k

Out: one digit per query

Constraints: $1 \leq q \leq 1000$, $1 \leq k \leq 10^{18}$.

3217 — Knight Moves Grid

A knight is on an $n \times n$ chessboard. For every square, determine the minimum number of knight moves needed to reach the top-left square. **In:** n

Out: $n \times n$ matrix of minimum move counts

Constraints: $4 \leq n \leq 1000$.

3311 — Grid Coloring I

An $n \times m$ grid contains A, B, C, or D. Change *every* cell to one of these four characters, different from its original character, so no horizontally or vertically adjacent cells are equal. **In:** $n \ m$; then n grid rows

Out: any valid final grid; if none exists, IMPOSSIBLE

Constraints: $1 \leq n, m \leq 500$.

3399 — Raab Game I

Each of two players has cards $1, 2, \dots, n$ and plays each card exactly once. Each round, the higher card scores one point; equal cards score neither. Given final scores a, b , construct a possible play order for both players. **In:** t ; then t cases $n \ a \ b$

Out: for each case, NO if impossible; otherwise YES and two permutations of $1, \dots, n$ realizing the scores

Constraints: $1 \leq t \leq 1000$, $1 \leq n \leq 100$, $0 \leq a, b \leq n$.

3419 — Mex Grid Construction

Construct an $n \times n$ grid. Each cell must contain the smallest nonnegative integer that appears neither to its left in the same row nor above it in the same column. **In:** n

Out: the required $n \times n$ grid

Constraints: $1 \leq n \leq 100$.

CSES Sorting and Searching — Problem Reference

All listed tasks: 1 s, 512 MB. Examples omitted for compactness.

1073 — Towers

Process n cubes in the given order. A cube may start a new tower or be placed on a tower whose current top is strictly larger. Minimize the number of towers. **In:** n ; then $k_1 \dots k_n$

Out: minimum number of towers

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq k_i \leq 10^9$.

1074 — Stick Lengths

Change every stick to one common length. Changing length p_i to x costs $|p_i - x|$. Minimize total cost. **In:** n ; then $p_1 \dots p_n$

Out: minimum total cost

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq p_i \leq 10^9$. Use 64-bit.

1085 — Array Division

Divide an array of positive integers into exactly k contiguous nonempty subarrays. Minimize the largest subarray sum. **In:** n k ; then $x_1 \dots x_n$

Out: minimum possible value of the maximum part sum

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

1091 — Concert Tickets

Tickets have prices h_i . Customers arrive in order with maximum prices t_i . Give each customer the most expensive remaining ticket with price $\leq t_i$; each ticket is sold at most once. **In:** n m ; prices h_i ; customer limits t_i

Out: price paid by each customer, or -1

Constraints: $1 \leq n, m \leq 2 \cdot 10^5$, $1 \leq h_i, t_i \leq 10^9$.

1141 — Playlist

Given a sequence of song IDs, find the longest contiguous segment in which every song ID is distinct. **In:** n ; then $k_1 \dots k_n$

Out: maximum length

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq k_i \leq 10^9$.

1163 — Traffic Lights

A street spans positions 0 through x . Insert n traffic lights one by one at distinct positions. After every insertion, find the longest interval containing no traffic light in its interior. **In:** x n ; then positions $p_1 \dots p_n$

Out: n longest passage lengths, after each insertion

Constraints: $1 \leq x \leq 10^9$, $n \leq 2 \cdot 10^5$, $0 < p_i < x$.

1164 — Room Allocation

Customer i stays from day a_i through b_i . A room can be reused only if the previous customer's departure day is *earlier* than the next arrival day. Minimize rooms and give an allocation. **In:** n ; then a b for each customer

Out: minimum room count k ; then each customer's room in input order

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq a \leq b \leq 10^9$.

1619 — Restaurant Customers

Given arrival and leaving times of n customers, find the maximum number simultaneously in the restaurant. All arrival/leaving times are distinct. **In:** n ; then a b for each customer

Out: maximum simultaneous customers

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq a < b \leq 10^9$.

1620 — Factory Machines

Machine i needs k_i seconds per product and machines work simultaneously. Find the minimum time needed to produce at least t products in total. **In:** n t ; then $k_1 \dots k_n$

Out: minimum required time

Constraints: $n \leq 2 \cdot 10^5$, $t \leq 10^9$, $k_i \leq 10^9$. Use 64-bit.

1629 — Movie Festival

Given start/end times of n movies, choose a maximum-size set that one person can watch completely without overlaps. **In:** n ; then a b for each movie

Out: maximum number of movies

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq a < b \leq 10^9$.

1630 — Tasks and Deadlines

Process every task exactly once, sequentially from time 0. Task i has duration a_i and deadline d_i ; if it finishes at f_i , its reward is $d_i - f_i$. Maximize total reward. **In:** n ; then a d for each task

Out: maximum total reward

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq a, d \leq 10^6$. Use 64-bit.

1631 — Reading Books

Two readers must each read all n books from beginning to end. Book i takes t_i time, and the two readers cannot read the same book simultaneously. Minimize the total finishing time. **In:** n ; then $t_1 \dots t_n$

Out: minimum total time

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq t_i \leq 10^9$. Use 64-bit.

1632 — Movie Festival II

There are k club members and n movies with start/end times. Each member may watch only non-overlapping movies. Maximize the total number watched across all members. **In:** n k ; then a b for each movie

Out: maximum total number of movies

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq a < b \leq 10^9$.

1640 — Sum of Two Values

Find two array values at distinct positions whose sum is exactly x . **In:** n x ; then $a_1 \dots a_n$

Out: two 1-indexed positions, or IMPOSSIBLE

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x, a_i \leq 10^9$.

1641 — Sum of Three Values

Find three array values at distinct positions whose sum is exactly x . **In:** n x ; then $a_1 \dots a_n$

Out: three 1-indexed positions, or IMPOSSIBLE

Constraints: $1 \leq n \leq 5000$, $1 \leq x, a_i \leq 10^9$.

1642 — Sum of Four Values

Find four array values at distinct positions whose sum is exactly x . **In:** n x ; then $a_1 \dots a_n$

Out: four 1-indexed positions, or IMPOSSIBLE

Constraints: $1 \leq n \leq 1000$, $1 \leq x, a_i \leq 10^9$.

1643 — Maximum Subarray Sum

Find the maximum sum of a contiguous, nonempty subarray. **In:** n ; then $x_1 \dots x_n$

Out: maximum subarray sum

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $-10^9 \leq x_i \leq 10^9$. Use 64-bit.

1644 — Maximum Subarray Sum II

Find the maximum sum of a contiguous subarray whose length is between a and b , inclusive. **In:** n a b ; then $x_1 \dots x_n$

Out: maximum valid subarray sum

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq a \leq b \leq n$, $|x_i| \leq 10^9$. Use 64-bit.

1645 — Nearest Smaller Values

For every array position i , find the nearest position $j < i$ with $x_j < x_i$. **In:** n ; then $x_1 \dots x_n$

Out: n positions; print 0 when no such j exists

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.

1660 — Subarray Sums I

Given an array of *positive* integers, count contiguous subarrays whose sum is exactly x . **In:** n x ; then $a_1 \dots a_n$

Out: number of matching subarrays

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x, a_i \leq 10^9$. Use 64-bit count.

1661 — Subarray Sums II

Given an integer array, count contiguous subarrays whose sum is exactly x . **In:** n x ; then $a_1 \dots a_n$

Out: number of matching subarrays

Constraints: $n \leq 2 \cdot 10^5$, $-10^9 \leq x, a_i \leq 10^9$. Use 64-bit.

1662 — Subarray Divisibility

Count contiguous subarrays whose sum is divisible by n . **In:** n ; then $a_1 \dots a_n$

Out: number of valid subarrays

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $-10^9 \leq a_i \leq 10^9$. Use 64-bit.

2162 — Josephus Problem I

Children $1, 2, \dots, n$ stand in a circle. Repeatedly remove every other remaining child until none remain.

In: n

Out: the complete removal order

Constraints: $1 \leq n \leq 2 \cdot 10^5$.

2163 — Josephus Problem II

Children $1, 2, \dots, n$ stand in a circle. Repeatedly skip k remaining children and remove the next child. **In:** n k

Out: the complete removal order

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $0 \leq k \leq 10^9$.

2168 — Nested Ranges Check

For each distinct range $[x, y]$, determine (1) whether it contains some other input range, and (2) whether some other input range contains it. $[a, b]$ contains $[c, d]$ iff $a \leq c$ and $d \leq b$. **In:** n ; then x y for each range

Out: two n -entry 0/1 lines, both in input order: contains; contained-by

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x < y \leq 10^9$; ranges distinct.

2169 — Nested Ranges Count

For each distinct range $[x, y]$, count (1) how many other ranges it contains and (2) how many other ranges contain it, using inclusive containment. **In:** n ; then x y for each range

Out: two n -entry lines in input order: contained count; containing count

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x < y \leq 10^9$; ranges distinct.

2183 — Missing Coin Sum

Given positive coin values, each usable at most once, find the smallest positive integer sum that cannot be formed by a subset. **In:** n ; then $x_1 \dots x_n$

Out: smallest unformable positive sum

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

2216 — Collecting Numbers

The array is a permutation of $1, \dots, n$. In each round scan left-to-right, collecting as many next required numbers as possible while collecting $1, 2, \dots, n$ in order. Find the number of rounds. **In:** n ; then permutation $x_1 \dots x_n$

Out: number of rounds

Constraints: $1 \leq n \leq 2 \cdot 10^5$.

2217 — Collecting Numbers II

Same collecting process as above. Perform m operations, each swapping the values at two positions a, b . Report the required number of rounds after every swap.

In: n m ; permutation; then m swaps a b

Out: m answers, one after each swap

Constraints: $1 \leq n, m \leq 2 \cdot 10^5$, $1 \leq a, b \leq n$.

2428 — Distinct Values Subarrays II

Count contiguous subarrays containing at most k distinct values. **In:** n k ; then $x_1 \dots x_n$

Out: number of valid subarrays

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

3420 — Distinct Values Subarrays

Count contiguous subarrays in which every element is distinct. **In:** n ; then $x_1 \dots x_n$

Out: number of all-distinct subarrays

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

3421 — Distinct Values Subsequences

Count nonempty subsequences in which every chosen element is distinct. Subsequences are determined by positions and may have gaps. Print the count modulo $10^9 + 7$. **In:** n ; then $x_1 \dots x_n$

Out: number of all-distinct subsequences mod $10^9 + 7$

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.

CSES Dynamic Programming — Problem Reference

All listed tasks: 1 s, 512 MB. Examples omitted for compactness.

1093 — Two Sets II

Count the ways to partition $\{1, 2, \dots, n\}$ into two sets having equal sum. The two sets are unordered: swapping the two sides is the same partition. **In:** n

Out: number of partitions modulo $10^9 + 7$

Constraints: $1 \leq n \leq 500$.

1097 — Removal Game

Two players alternately remove the first or last number of a list and add it to their score. Both play optimally. Find the maximum final score the first player can guarantee. **In:** n ; then $x_1 \dots x_n$

Out: maximum score of the first player

Constraints: $1 \leq n \leq 5000$, $-10^9 \leq x_i \leq 10^9$. Use 64-bit.

1140 — Projects

Project i occupies every day from a_i through b_i and gives reward p_i . Choose non-overlapping projects, with at most one attended on any day, maximizing total reward. **In:** n ; then $a_1 \dots b_1 \dots p_1$

Out: maximum total reward

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq a_i \leq b_i \leq 10^9$, $p_i \leq 10^9$. Use 64-bit.

1145 — Increasing Subsequence

Given an integer array, find the length of its longest *strictly increasing* subsequence. A subsequence preserves order but may skip elements. **In:** n ; then $x_1 \dots x_n$

Out: LIS length

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.

1158 — Book Shop

There are n books; book i costs h_i and has s_i pages. Buy each book at most once, spending at most x , and maximize total pages. **In:** n x ; prices h_i ; pages s_i

Out: maximum number of pages

Constraints: $n \leq 1000$, $x \leq 10^5$, $1 \leq h_i, s_i \leq 1000$.

1633 — Dice Combinations

Count ordered sequences of dice throws from $\{1, 2, \dots, 6\}$ whose sum is exactly n . **In:** n

Out: number of sequences modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^6$.

1634 — Minimizing Coins

Given distinct positive coin values, each usable any number of times, form sum x using the minimum number of coins. **In:** n x ; then $c_1 \dots c_n$

Out: minimum coin count, or -1 if impossible

Constraints: $n \leq 100$, $x \leq 10^6$, $1 \leq c_i \leq 10^6$.

1635 — Coin Combinations I

Given distinct positive coin values, each reusable, count *ordered* sequences of coins whose values sum to x . Different orders count separately. **In:** n x ; then $c_1 \dots c_n$

Out: number of ordered combinations modulo $10^9 + 7$

Constraints: $n \leq 100$, $x \leq 10^6$, $1 \leq c_i \leq 10^6$.

1636 — Coin Combinations II

Given distinct positive coin values, each reusable, count combinations whose values sum to x when order does *not* matter. **In:** n x ; then $c_1 \dots c_n$

Out: number of combinations modulo $10^9 + 7$

Constraints: $n \leq 100$, $x \leq 10^6$, $1 \leq c_i \leq 10^6$.

1637 — Removing Digits

Starting from n , each step subtracts one decimal digit currently appearing in the number. Find the minimum steps needed to reach 0. **In:** n

Out: minimum number of steps

Constraints: $1 \leq n \leq 10^6$.

1638 — Grid Paths I

An $n \times n$ grid contains empty cells $.$ and traps $*$. Move from the upper-left to the lower-right using only right/down moves and never enter a trap. **In:** n ; then n grid rows

Out: number of valid paths modulo $10^9 + 7$

Constraints: $1 \leq n \leq 1000$.

1639 — Edit Distance

Find the minimum number of single-character insertions, deletions, and replacements needed to transform one uppercase string into the other. **In:** first string; second string

Out: edit distance

Constraints: $1 \leq n, m \leq 5000$.

1653 — Elevator Rides

There are n people with weights w_i and an elevator capacity x . Every person must reach the top; each ride may carry any subset whose total weight is at most x . Minimize rides. **In:** n x ; then $w_1 \dots w_n$

Out: minimum number of rides

Constraints: $1 \leq n \leq 20$, $1 \leq w_i \leq x \leq 10^9$.

1744 — Rectangle Cutting

Cut an $a \times b$ integer-sided rectangle into squares. Each move cuts one current rectangle into two rectangles along an integer coordinate. Minimize the number of cuts. **In:** a b

Out: minimum number of cuts

Constraints: $1 \leq a, b \leq 500$.

1745 — Money Sums

Given n positive-value coins, each usable at most once, find every positive sum obtainable from a subset. **In:** n ; then $x_1 \dots x_n$

Out: number k of distinct positive sums; then all sums in increasing order

Constraints: $1 \leq n \leq 100$, $1 \leq x_i \leq 1000$.

1746 — Array Description

Count arrays of length n with values in $[1, m]$ such that adjacent values differ by at most 1. Input positions may be fixed; value 0 means unknown. **In:** n m ; then $x_1 \dots x_n$

Out: number of matching arrays modulo $10^9 + 7$

Constraints: $n \leq 10^5$, $m \leq 100$, $0 \leq x_i \leq m$.

1748 — Increasing Subsequence II

Count all nonempty *strictly increasing* subsequences of an integer array. Subsequences using different positions are distinct even if their values match. **In:** n ; then $x_1 \dots x_n$

Out: number of increasing subsequences modulo $10^9 + 7$

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.

2181 — Counting Tilings

Count ways to tile an $n \times m$ rectangle completely using 1×2 and 2×1 dominoes. **In:** n m

Out: number of tilings modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10$, $1 \leq m \leq 1000$.

2220 — Counting Numbers

Count integers in the inclusive range $[a, b]$ whose decimal representation has no two equal adjacent digits. **In:** a b

Out: number of valid integers

Constraints: $0 \leq a \leq b \leq 10^{18}$. Use 64-bit.

2413 — Counting Towers

For each query n , count different towers of width 2 and height n built from integer-sided blocks. Distinct-looking mirrored or rotated constructions count separately. **In:** t ; then one n per test

Out: answer for each test modulo $10^9 + 7$

Constraints: $1 \leq t \leq 100$, $1 \leq n \leq 10^6$.

3314 — Mountain Range

Mountains stand in a row. You may glide from mountain a to b when a is taller than b and taller than every mountain strictly between them. Start anywhere. Maximize mountains visited. **In:** n ; then $h_1 \dots h_n$

Out: maximum number of visited mountains

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq h_i \leq 10^9$.

3359 — Minimal Grid Path

An $n \times n$ grid contains uppercase letters. Move from the upper-left to the lower-right using only right/down moves; concatenate every visited cell. Find the lexicographically smallest possible string. **In:** n ; then n rows of n letters

Out: lexicographically minimum path string

Constraints: $1 \leq n \leq 3000$.

3403 — Longest Common Subsequence

Given two integer arrays, find a longest sequence obtainable as a subsequence of both. A subsequence preserves order but may skip elements. **In:** n m ; array a ; array b

Out: LCS length; then any one longest common subsequence

Constraints: $1 \leq n, m \leq 1000$, $1 \leq a_i, b_i \leq 10^9$.

CSES Graph Algorithms — Problem Reference

All listed tasks: 1 s; 512 MB unless noted. Examples omitted for compactness.

1160 — Planets Queries II

Each planet has exactly one outgoing teleporter t_i . For each query (a, b) , find the minimum number of teleportations needed to reach b from a , or determine that it is impossible. **In:** n q ; destinations $t_1 \dots t_n$; then a b queries

Out: minimum jumps for each query, or -1
Constraints: $1 \leq n, q \leq 2 \cdot 10^5$, $1 \leq t_i, a, b \leq n$.

1192 — Counting Rooms

An $n \times m$ map contains floor $.$ and walls $\#$. Moving in four directions through floor cells, count the connected floor components (rooms).

In: n m ; then n grid rows

Out: number of rooms

Constraints: $1 \leq n, m \leq 1000$.

1193 — Labyrinth

In a grid with walls $\#$, floor $.$, one start A , and one target B , find a shortest four-direction path from A to B . **In:** n m ; then n grid rows
Out: NO, or YES, path length, and any shortest path using L/R/U/D

Constraints: $1 \leq n, m \leq 1000$.

1194 — Monsters

You and all monsters move simultaneously one grid step per turn. Escape from A to any boundary cell without ever occupying a cell at the same time as a monster; the route must be safe even if monsters know it. **In:** n m ; grid with $.$, $\#$, one A , and zero or more M

Out: NO, or YES, a valid path length and D/U/L/R path

Constraints: $1 \leq n, m \leq 1000$; printed path length $\leq nm$.

1195 — Flight Discount

Directed flights have positive prices. Travel from city 1 to city n and use one coupon on a single edge, changing price c to $\lfloor c/2 \rfloor$. Minimize total cost. **In:** n m ; then directed edges a b c

Out: minimum route price

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $c \leq 10^9$; a route exists. Use 64-bit.

1196 — Flight Routes

Find the k cheapest directed routes from city 1 to city n . Routes may revisit vertices, equal-cost routes count separately, and at least k routes exist. **In:** n m k ; then directed edges a b c

Out: k route costs in nondecreasing order

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $c \leq 10^9$, $1 \leq k \leq 10$. Use 64-bit.

1197 — Cycle Finding

Given a directed weighted graph, determine whether it contains a cycle whose total edge weight is negative, and reconstruct one if it exists. **In:** n m ; then directed edges a b c

Out: NO, or YES followed by the cycle vertices in order (start repeated at end)

Constraints: $n \leq 2500$, $m \leq 5000$, $|c| \leq 10^9$. Use 64-bit.

1202 — Investigation

For directed positive-cost flights from 1 to n , determine: minimum route price, number of minimum-price routes modulo $10^9 + 7$, minimum flights among them, and maximum flights among them. **In:** n m ; then directed edges a b c

Out: four integers: distance, count, min edges, max edges

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $c \leq 10^9$; a route exists. Use 64-bit.

1666 — Building Roads

Given an undirected graph, add the minimum number of roads so that every pair of cities becomes connected. Output any set of roads achieving this. **In:** n m ; then undirected edges a b

Out: number k of added roads; then k road endpoints

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$; no loops or duplicate roads.

1667 — Message Route

In an undirected unweighted network, find a route from computer 1 to n using the minimum number of computers (equivalently, shortest number of edges) and reconstruct it. **In:** n m ; then undirected edges a b

Out: IMPOSSIBLE, or route vertex count and any shortest route

Constraints: $2 \leq n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1668 — Building Teams

Split the vertices of an undirected friendship graph into two teams so every edge has endpoints in different teams. **In:** n m ; then undirected edges a b

Out: a label 1 or 2 for each vertex, or IMPOSSIBLE

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1669 — Round Trip

Find any simple cycle in an undirected graph: the route must start/end at the same city and all intermediate cities must be distinct. **In:** n m ; then undirected edges a b

Out: IMPOSSIBLE, or cycle vertex count and vertices in order

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$; no loops or parallel edges.

1671 — Shortest Routes I

In a directed graph with positive edge lengths, find the shortest distance from city 1 to every city. All cities are reachable from 1. **In:** n m ; then directed edges a b c

Out: n distances from city 1

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $c \leq 10^9$. Use 64-bit.

1672 — Shortest Routes II

Given an undirected weighted graph and many queries, answer the shortest distance between each queried pair of cities; pairs may be disconnected. **In:** n m q ; m undirected edges a b c ; then q queries a b

Out: distance per query, or -1 if unreachable

Constraints: $n \leq 500$, $m \leq n^2$, $q \leq 10^5$, $c \leq 10^9$. Use 64-bit.

1673 — High Score

Directed edges may add positive or negative score. Starting at 1, reach n with maximum total score; edges may be reused. If the score can be made arbitrarily large on some valid $1 \rightarrow n$ route, print -1 . **In:** n m ; then directed edges a b x

Out: maximum score, or -1 if unbounded

Constraints: $n \leq 2500$, $m \leq 5000$, $|x| \leq 10^9$; n is reachable from 1. Use 64-bit.

1675 — Road Reparation

Undirected roads have repair costs. Choose roads of minimum total cost so the repaired network is connected; if impossible, report it. **In:** n m ; then undirected roads a b c

Out: minimum total repair cost, or IMPOSSIBLE

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $c \leq 10^9$. Memory 128 MB. Use 64-bit.

1676 — Road Construction

Start with n isolated cities and add m undirected roads one by one. After each addition, report the current number of connected components and the size of the largest component.

In: n m ; then roads a b in insertion order

Out: after each road: component count and largest component size

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1678 — Round Trip II

Find any directed cycle: start/end at the same city and all intermediate cities must be distinct. **In:** n m ; then directed edges a b

Out: IMPOSSIBLE, or cycle vertex count and vertices in order

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1679 — Course Schedule

Given prerequisites $a \rightarrow b$ meaning course a must come before b , output any ordering containing every course that satisfies all requirements. **In:** n m ; then prerequisite edges a b

Out: a valid course order, or IMPOSSIBLE

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1680 — Longest Flight Route

The directed flight graph is a DAG. Find a route from city 1 to city n that visits the maximum possible number of cities, and reconstruct it. **In:** n m ; then directed edges a b

Out: IMPOSSIBLE, or maximum city count and any maximizing route

Constraints: $2 \leq n \leq 10^5$, $m \leq 2 \cdot 10^5$; graph is acyclic.

1681 — Game Routes

A directed acyclic graph has teleporters between levels. Count the number of directed routes from level 1 to level n . **In:** n m ; then directed edges a b

Out: number of routes modulo $10^9 + 7$

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$; graph is acyclic.

1682 — Flight Routes Check

Determine whether a directed graph is strongly connected, i.e. every city can reach every other city. **In:** n m ; then directed edges a b

Out: YES, or NO and any pair a b such that a cannot reach b

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1683 — Planets and Kingdoms

Partition a directed graph into strongly connected components: two planets share a kingdom exactly when each can reach the other.

In: n m ; then directed edges a b

Out: number k of SCCs; then one kingdom label $1..k$ for each planet

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1684 — Giant Pizza

There are m Boolean toppings and n clauses. Each clause contains two literals $+x$ or $-x$; choose topping truth values so at least one literal in every clause is true. **In:** n m ; then n pairs of signed topping literals

Out: one $+/-$ choice for each topping, or IMPOSSIBLE

Constraints: $1 \leq n, m \leq 10^5$, topping indices $1..m$.

1686 — Coin Collector

Each room has k_i coins and directed tunnels connect rooms. You may start and end anywhere and move along tunnels; maximize the total coins obtainable from visited rooms. **In:** n m ; coin values $k_1 \dots k_n$; then directed edges a b

Out: maximum collectible coins

Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $k_i \leq 10^9$. Use 64-bit.

1689 — Knight's Tour

Given a starting square on an 8×8 board, construct any knight's tour that visits every square exactly once. **In:** x y , the starting position

Out: an 8×8 grid whose entries $1..64$ give the visit order

Constraints: $1 \leq x, y \leq 8$.

1690 — Hamiltonian Flights

Count directed routes from city 1 to city n that visit every city exactly once. **In:** n m ; then directed edges a b

Out: number of Hamiltonian $1 \rightarrow n$ routes modulo $10^9 + 7$

Constraints: $2 \leq n \leq 20$, $1 \leq m \leq n^2$.

1691 — Mail Delivery

In an undirected graph, find a closed route starting and ending at crossing 1 that uses every street exactly once. **In:** n m ; then undirected edges a b

Out: an Euler circuit as $m + 1$ crossings, or IMPOSSIBLE

Constraints: $2 \leq n \leq 10^5$, $m \leq 2 \cdot 10^5$; no loops or parallel streets.

1692 — De Bruijn Sequence

Construct a minimum-length binary string containing every possible binary string of length n as a contiguous substring. **In:** n

Out: any minimum-length valid bit string

Constraints: $1 \leq n \leq 15$.

1693 — Teleporters Path

In a directed graph, start at level 1, finish at level n , and use every teleporter (edge) exactly once. **In:** n m ; then distinct directed edges a b

Out: an Euler trail as $m + 1$ vertices, or IMPOSSIBLE

Constraints: $2 \leq n \leq 10^5$, $m \leq 2 \cdot 10^5$.

1694 — Download Speed

A directed network edge $a \rightarrow b$ has capacity c . Find the maximum total data flow from server 1 to computer n . **In:** n m ; then directed capacities a b c

Out: maximum flow value

Constraints: $n \leq 500$, $m \leq 1000$, $c \leq 10^9$. Use 64-bit.

1695 — Police Chase

An undirected street graph connects bank 1 to harbor n . Find the minimum number of streets whose removal disconnects them, and output one such minimum cut. **In:** n m ; then undirected streets a b

Out: cut size k ; then the k streets to close

Constraints: $2 \leq n \leq 500$, $m \leq 1000$; at most one street per pair.

1696 — School Dance

Given a bipartite graph of n boys, m girls, and k acceptable pairs, find a maximum matching and output the matched pairs. **In:** n m k ; then acceptable pairs boy girl

Out: matching size r ; then r matched pairs

Constraints: $n, m \leq 500$, $k \leq 1000$.

1711 — Distinct Routes

Directed teleporters may each be used on at most one day. Every day starts at room 1 and ends at room n . Maximize the number of edge-disjoint $1 \rightarrow n$ routes and output them. **In:** n m ; then distinct directed edges a b

Out: maximum route count k ; then for each route its vertex count and vertices

Constraints: $2 \leq n \leq 500$, $m \leq 1000$.

1750 — Planets Queries I

Each planet has one outgoing teleporter t_i . For each query (x, k) , determine the planet reached after exactly k teleportations from x . **In:** n q ; destinations $t_1 \dots t_n$; then queries x k

Out: destination planet for each query

Constraints: $n, q \leq 2 \cdot 10^5$, $0 \leq k \leq 10^9$.

1751 — Planets Cycles

Each planet has one outgoing teleporter. Starting from each planet, keep teleporting until the next planet has already been visited in that walk. Report the number of teleportations performed. **In:** n ; destinations $t_1 \dots t_n$

Out: n answers, one for each starting planet

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq t_i \leq n$.

CSES — Range Queries Problem Reference

All tasks: 1 s, 512 MB. Ranges and array indices are 1-based.

1143 — Hotel Queries

n hotels have free-room counts h_i . For each arriving group requiring r , assign the *first* hotel with at least r rooms, then decrease that hotel's free rooms by r ; output 0 if none exists. **In:** n m ; $h_1 \dots h_n$; $r_1 \dots r_m$
Out: assigned hotel index for every group, or 0
Constraints: $n, m \leq 2 \cdot 10^5$; $h_i, r_i \leq 10^9$.

1144 — Salary Queries

Maintain salaries p_i under point assignments. Query how many employees have salary in the inclusive value interval $[a, b]$. **In:** n q ; salaries; then $!$ k x or $?$ a b
Out: count for each $?$ query
Constraints: $n, q \leq 2 \cdot 10^5$; salaries/query values $\leq 10^9$.

1190 — Subarray Sum Queries

Maintain an integer array under point assignments. After *every* update, report the maximum subarray sum of the entire array; the empty subarray with sum 0 is allowed. **In:** n m ; array; then m updates k x
Out: maximum subarray sum after each update
Constraints: $n, m \leq 2 \cdot 10^5$; $|x_i|, |x| \leq 10^9$. Use 64-bit.

1646 — Static Range Sum Queries

Given a fixed array, answer the sum of each inclusive range $[a, b]$. **In:** n q ; array; then q lines a b
Out: $\sum_{i=a}^b x_i$ for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$. Use 64-bit.

1647 — Static Range Minimum Queries

Given a fixed array, answer the minimum value in each inclusive range $[a, b]$. **In:** n q ; array; then q lines a b
Out: $\min(x_a, \dots, x_b)$ for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$.

1648 — Dynamic Range Sum Queries

Maintain an array with point assignment and range-sum queries. **In:** n q ; array; queries 1 k u (set $x_k = u$) or 2 a b (sum)
Out: range sum for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i, u \leq 10^9$. Use 64-bit.

1649 — Dynamic Range Minimum Queries

Maintain an array with point assignment and range-minimum queries. **In:** n q ; array; queries 1 k u (set) or 2 a b (minimum)
Out: range minimum for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i, u \leq 10^9$.

1650 — Range Xor Queries

Given a fixed array, answer the bitwise xor of values in each inclusive range $[a, b]$. **In:** n q ; array; then q lines a b
Out: $x_a \oplus \dots \oplus x_b$ for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$.

1651 — Range Update Queries

Maintain an array under range additions. Type 1 adds u to every x_i for $i \in [a, b]$; type 2 asks the current value at one position k . **In:** n q ; array; 1 a b u or 2 k
Out: x_k for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i, u \leq 10^9$. Use 64-bit.

1652 — Forest Queries

An $n \times n$ fixed grid contains trees $*$ and empty cells $.$. For each rectangle with corners (y_1, x_1) and (y_2, x_2) , count trees inside it, boundaries included. **In:** n q ; n grid rows; then y_1 x_1 y_2 x_2
Out: tree count for each rectangle
Constraints: $n \leq 1000$, $q \leq 2 \cdot 10^5$.

1664 — Movie Festival Queries

n movies have intervals $[s, e]$. For each arrival/leaving interval $[a, b]$, find the maximum number of whole movies that can be watched: first movie may start at a , consecutive movies may touch ($e_1 \leq s_2$), and last may end at b . **In:** n q ; n movie lines a b ; then q availability lines a b
Out: maximum watchable movies for each query
Constraints: $n, q \leq 2 \cdot 10^5$; all times in $[1, 10^6]$, start $<$ end.

1734 — Distinct Values Queries

Given a fixed array, count the number of distinct values in each range $[a, b]$. **In:** n q ; array; then q lines a b
Out: number of distinct values in each range
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$.

1735 — Range Updates and Sums

Maintain an array with three operations: add x to all values in $[a, b]$; set all values in $[a, b]$ to x ; or query the sum of $[a, b]$. **In:** n q ; array; then 1 a b x , 2 a b x , or 3 a b
Out: range sum for each type-3 query
Constraints: $n, q \leq 2 \cdot 10^5$; initial/update values $\leq 10^6$. Use 64-bit.

1736 — Polynomial Queries

Type 1 on $[a, b]$ adds $1, 2, \dots, b - a + 1$ respectively to $x_a, x_{a+1}, \dots, x_b$. Type 2 asks the sum of $[a, b]$. **In:** n q ; array; then 1 a b or 2 a b
Out: range sum for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^6$. Use 64-bit.

1737 — Range Queries and Copies

Maintain a list of versioned arrays, initially containing one array. Operations: set position a in version k to x ; query sum $[a, b]$ in version k ; or append an independent copy of version k as a new version. **In:** n q ; array; then 1 k a x , 2 k a b , or 3 k
Out: range sum for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^3$; $t_i, x \leq 10^9$. Use 64-bit.

1739 — Forest Queries II

Maintain an $n \times n$ tree grid. Type 1 toggles cell (y, x) between empty/tree; type 2 counts trees in an inclusive query rectangle. **In:** n q ; grid; then 1 y x or 2 y_1 x_1 y_2 x_2
Out: tree count for each type-2 query
Constraints: $n \leq 1000$, $q \leq 2 \cdot 10^5$.

1749 — List Removals

Start with list $x_1, \dots, x_n$. Repeatedly remove and print the element currently at position p_i ; positions are relative to the shrinking list. **In:** n ; values $x_1 \dots x_n$; removals $p_1 \dots p_n$
Out: removed values in order
Constraints: $n \leq 2 \cdot 10^3$; $x_i \leq 10^9$; $1 \leq p_i \leq n - i + 1$.

2166 — Prefix Sum Queries

Maintain an array under point assignments. For a query range $[a, b]$, report the maximum prefix sum $\max(0, x_a, x_a + x_{a+1}, \dots, \sum_{i=a}^b x_i)$. **In:** n q ; array; 1 k u (set) or 2 a b (query)
Out: maximum prefix sum for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $|x_i|, |u| \leq 10^9$. Use 64-bit.

2184 — Missing Coin Sum Queries

Coins have positive values x_i . For each index range $[a, b]$, using each coin there at most once, find the smallest positive sum that cannot be formed. **In:** n q ; coin values; then q lines a b
Out: smallest nonrepresentable positive sum for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$. Use 64-bit.

2206 — Pizzeria Queries

Building i sells pizza for p_i . Delivery from pizzeria a to customer at k costs $p_a + |a - k|$. Support point price changes and queries for the minimum delivered price at building k . **In:** n q ; prices; then 1 k x or 2 k
Out: $\min_a (p_a + |a - k|)$ for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $p_i, x \leq 10^9$. Use 64-bit.

2416 — Increasing Array Queries

For each subarray $x_a, \dots, x_b$, you may only increment elements. Find the minimum total number of $+1$ operations needed to make it nondecreasing ($x_i \geq x_{i-1}$). **In:** n q ; array; then q lines a b
Out: minimum increments for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$. Use 64-bit.

3163 — Range Interval Queries

Given fixed $x_1, \dots, x_n$, each query (a, b, c, d) asks how many indices satisfy both $a \leq i \leq b$ and $c \leq x_i \leq d$. **In:** n q ; array; then a b c d
Out: matching index count for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i, c, d \leq 10^9$.

3226 — Subarray Sum Queries II

Given a fixed integer array, for each range $[a, b]$ report the maximum subarray sum contained entirely inside that range. Empty subarrays with sum 0 are allowed. **In:** n q ; array; then q lines a b
Out: maximum subarray sum in each query range
Constraints: $n, q \leq 2 \cdot 10^5$; $|x_i| \leq 10^9$. Use 64-bit.

3304 — Visible Buildings Queries

For each query $[a, b]$, pretend only those buildings exist and view them from the left. Count buildings whose height is *strictly greater* than every earlier height in the range. **In:** n q ; heights; then q lines a b
Out: number of visible buildings per query
Constraints: $n \leq 10^5$, $q \leq 2 \cdot 10^5$; $h_i \leq 10^9$.

3356 — Distinct Values Queries II

Maintain an array under point assignments. For each range query $[a, b]$, determine whether every value in the range is pairwise distinct. **In:** n q ; array; then 1 k u or 2 a b
Out: YES if all values in $[a, b]$ are distinct, otherwise NO
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i, u \leq 10^9$.

CSES – Mathematics Problem Reference

All tasks: 1 s, 512 MB. Arithmetic is modulo $10^9 + 7$ only where explicitly stated.

1079 – Binomial Coefficients

Compute n values $\binom{a}{b}$ modulo $10^9 + 7$, where $0 \leq b \leq a$. **In:** n ; then n lines **a b**

Out: each $\binom{a}{b} \bmod 10^9 + 7$

Constraints: $n \leq 10^5$; $a \leq 10^6$.

1081 – Common Divisors

Given n positive integers, find two array elements whose greatest common divisor is as large as possible. **In:** n ; **x_1**

... **x_n**

Out: maximum possible gcd

Constraints: $2 \leq n \leq 2 \cdot 10^5$; $x_i \leq 10^6$.

1082 – Sum of Divisors

Let $\sigma(i)$ be the sum of divisors of i . Compute $\sum_{i=1}^n \sigma(i) \bmod 10^9 + 7$. **In:** n

Out: the sum modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^{12}$.

1095 – Exponentiation

For each query, compute $a^b \bmod 10^9 + 7$. By definition here, $0^0 = 1$. **In:** n ; then n lines **a b**

Out: one modular power per query

Constraints: $n \leq 2 \cdot 10^5$; $0 \leq a, b \leq 10^9$.

1096 – Throwing Dice

Count ordered sequences of dice throws, each in $\{1, \dots, 6\}$, whose total sum is exactly n . **In:** n

Out: number of ways modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^{18}$.

1098 – Nim Game II

There are n heaps. A move chooses one nonempty heap and removes 1, 2, or 3 sticks. Last move wins; determine the winner under optimal play. **In:** t ; for each test: n ; heap sizes

Out: **first** or **second** for each test

Constraints: $t, n \leq 2 \cdot 10^5$; $x_i \leq 10^9$; total $n \leq 2 \cdot 10^5$.

1099 – Stair Game

Stairs $1 \dots n$ contain balls. A move chooses $k \neq 1$ with balls and moves any positive number of balls from stair k to $k - 1$. Last move wins; with no legal move initially, second wins.

In: t ; for each test: n ; **p_1** **...** **p_n**

Out: **first** or **second**

Constraints: $t, n \leq 2 \cdot 10^5$; $0 \leq p_i \leq 10^9$; total $n \leq 2 \cdot 10^5$.

1712 – Exponentiation II

For each query, compute $a^{b^c} \bmod 10^9 + 7$. By definition here, $0^0 = 1$. **In:** n ; then n lines **a b c**

Out: one answer per query

Constraints: $n \leq 10^5$; $0 \leq a, b, c \leq 10^9$.

1713 – Counting Divisors

For each given integer x , report the number of positive divisors of x . **In:** n ; then n values x

Out: divisor count for each x

Constraints: $n \leq 10^5$; $x \leq 10^6$.

1715 – Creating Strings II

Given a lowercase string, count the number of distinct strings obtainable by permuting all of its characters. **In:** one lowercase string of length n

Out: count modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^6$.

1716 – Distributing Apples

Distribute m identical apples among n children; each child may receive any nonnegative number. Count all distributions.

In: n m

Out: number of distributions modulo $10^9 + 7$

Constraints: $1 \leq n, m \leq 10^6$.

1717 – Christmas Party

n children each brought one gift. Count assignments of gifts to children such that nobody receives their own gift. **In:** n

Out: number of valid distributions modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^6$.

1722 – Fibonacci Numbers

With $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-2} + F_{n-1}$, compute F_n . **In:** n

Out: $F_n \bmod 10^9 + 7$

Constraints: $0 \leq n \leq 10^{18}$.

1723 – Graph Paths I

Given a directed graph, count paths from node 1 to node n that use exactly k edges. **In:** n m k ; then m directed edges **a b**

Out: number of length- k paths modulo $10^9 + 7$

Constraints: $n \leq 100$; $m \leq n(n - 1)$; $k \leq 10^9$.

1724 – Graph Paths II

Given a directed weighted graph, find the minimum total weight of a path from node 1 to node n using exactly k edges.

In: n m k ; then edges **a b c**

Out: minimum length, or -1 if no such path exists

Constraints: $n \leq 100$; $m \leq n(n - 1)$; $k \leq 10^9$; $c \leq 10^9$.

1725 – Dice Probability

Throw a fair six-sided die n times. Find the probability that the total sum lies in $[a, b]$. **In:** n **a b**

Out: probability rounded to 6 decimals

Constraints: $n \leq 100$; $1 \leq a \leq b \leq 6n$.

1726 – Moving Robots

Initially every square of an 8×8 board has one robot. Each robot independently makes k random legal orthogonal moves, choosing uniformly among directions that stay on the board. Find the expected number of empty squares afterward. **In:** k

Out: expected empty squares, rounded to 6 decimals

Constraints: $1 \leq k \leq 100$.

1727 – Candy Lottery

n children independently receive a uniformly random integer number of candies from 1 to k . Find the expected maximum received by any child. **In:** n k

Out: expected maximum, rounded to 6 decimals

Constraints: $1 \leq n, k \leq 100$.

1728 – Inversion Probability

For each position i , choose x_i independently and uniformly from $1 \dots r_i$. Find the expected number of inversions (a, b) with $a < b$ and $x_a > x_b$. **In:** n ; **r_1** **...** **r_n**

Out: expected inversions, rounded to 6 decimals

Constraints: $n \leq 100$; $r_i \leq 100$.

1729 – Stick Game

A heap has s sticks. Allowed removals are the distinct values in P , and $1 \in P$. For every initial size $s = 1, \dots, n$, classify the position as winning or losing under optimal play; last move wins. **In:** n k ; $p_1 \dots p_k$

Out: length- n string of W/L

Constraints: $n \leq 10^6$; $k \leq 100$; $p_i \leq n$.

1730 – Nim Game I

There are n heaps. A move chooses one nonempty heap and removes any positive number of sticks. Last move wins; determine the winner under optimal play. **In:** t ; for each test: n ; heap sizes

Out: first or second

Constraints: $t, n \leq 2 \cdot 10^5$; $x_i \leq 10^9$; total $n \leq 2 \cdot 10^5$.

2064 – Bracket Sequences I

Count valid parenthesis sequences of total length n . **In:** n

Out: number of sequences modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^6$.

2164 – Josephus Queries

Children $1 \dots n$ stand in a circle; repeatedly remove every second remaining child until none remain. For each query (n, k) , report the k -th child removed. **In:** q ; then q lines n k

Out: k -th removed child for each query

Constraints: $q \leq 10^5$; $1 \leq k \leq n \leq 10^9$.

2182 – Divisor Analysis

A positive integer is given by its prime factorization $\prod x_i^{k_i}$, with distinct primes x_i . Compute the number, sum, and product of all positive divisors. **In:** n ; then n lines x k

Out: divisor count, divisor sum, divisor product; all modulo $10^9 + 7$

Constraints: $n \leq 10^5$; $x \leq 10^6$; $k \leq 10^9$.

2185 – Prime Multiples

Given k distinct primes a_i and integer n , count integers in $1 \dots n$ divisible by at least one a_i . **In:** n k ; $a_1 \dots a_k$

Out: number of divisible integers

Constraints: $n \leq 10^{18}$; $k \leq 20$; $2 \leq a_i \leq n$.

2187 – Bracket Sequences II

A prefix of a bracket sequence is fixed. Count valid bracket sequences of total length n that begin with exactly this prefix.

In: n ; prefix string of length k

Out: number of completions modulo $10^9 + 7$

Constraints: $1 \leq k \leq n \leq 10^6$.

2207 – Grundy's Game

Start with one heap of n coins. A move chooses a heap and splits it into two nonempty heaps of different sizes. Last move wins; determine the winner under optimal play. **In:** t ; then t values n

Out: first or second for each test

Constraints: $t \leq 10^5$; $n \leq 10^6$.

2208 – Another Game

There are n nonempty heaps. A move selects some of the currently nonempty heaps and removes one coin from each selected heap. Last move wins; determine the winner. **In:** t ; for each test: n ; heap sizes

Out: first or second

Constraints: $t, n \leq 2 \cdot 10^5$; $x_i \leq 10^9$; total $n \leq 2 \cdot 10^5$.

2209 – Counting Necklaces

Count colorings of a necklace with n pearls and m possible colors, where colorings equivalent by rotation are considered the same. **In:** n m

Out: number of distinct necklaces modulo $10^9 + 7$

Constraints: $1 \leq n, m \leq 10^6$.

2210 – Counting Grids

Count black/white colorings of an $n \times n$ grid, considering two grids equal if one can be rotated to obtain the other. **In:** n

Out: number of distinct grids modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^9$.

2417 – Counting Coprime Pairs

Given n positive integers, count unordered index pairs whose values are coprime, i.e. have gcd 1. **In:** n ; $x_1 \dots x_n$

Out: number of coprime pairs

Constraints: $n \leq 10^5$; $x_i \leq 10^6$.

3154 – System of Linear Equations

Solve n linear equations in m variables over the field modulo $10^9 + 7$: $\sum_j a_{i,j} x_j \equiv b_i \pmod{10^9 + 7}$. Any valid solution is accepted. **In:** n m ; then n rows containing m coefficients and b_i

Out: $x_1, \dots, x_m$ with $0 \leq x_i < 10^9 + 7$, or -1 if impossible

Constraints: $n, m \leq 500$; all coefficients and b_i are in $[0, 10^9 + 7)$.

3355 – Sum of Four Squares

For each nonnegative integer n , find nonnegative integers a, b, c, d such that $n = a^2 + b^2 + c^2 + d^2$. **In:** t ; then t values n

Out: one valid quadruple per test

Constraints: $t \leq 1000$; $n \leq 10^7$; sum of all $n \leq 10^7$.

3396 – Next Prime

For each positive integer n , find the smallest prime strictly greater than n . **In:** t ; then t values n

Out: next prime after each n

Constraints: $t \leq 20$; $n \leq 10^{12}$.

3397 – Permutation Order

Let $p(n, k)$ be the k -th permutation of $1 \dots n$ in lexicographic order. Support both directions: given (n, k) output $p(n, k)$, or given n and the permutation output its rank k . **In:** t ; each test is 1 n k or 2 n $p_1 \dots p_n$

Out: the requested permutation or rank

Constraints: $t \leq 1000$; $n \leq 20$; $1 \leq k \leq n!$.

3398 – Permutation Rounds

Start from sorted array $[1, 2, \dots, n]$. In each round, the element currently at position i moves to position p_i . Find the first positive number of rounds after which the array is sorted again. **In:** n ; permutation $p_1 \dots p_n$

Out: number of rounds modulo $10^9 + 7$

Constraints: $1 \leq n \leq 2 \cdot 10^5$.

3406 – Triangle Number Sums

Triangle numbers are $1 + 2 + \dots + k$ for $k \geq 1$. For each positive integer n , find the minimum number of triangle numbers whose sum is n . **In:** t ; then t values n

Out: minimum count for each test

Constraints: $t \leq 100$; $n \leq 10^{12}$.

CSES — String Algorithms Problem Reference

All tasks: 1 s, 512 MiB. Counts are modulo $10^9 + 7$ where stated.

1731 — Word Combinations

Given lowercase string s and a dictionary of k distinct words, count the ordered ways to split all of s into dictionary words (words may be reused). **In:** s ; k ; then k words

Out: number of complete segmentations mod $10^9 + 7$

Constraints: $|s| \leq 5000$, $k \leq 10^5$; total dictionary length $\leq 10^6$.

1753 — String Matching

Given lowercase text s and pattern p , count starting positions where p occurs in s ; overlapping occurrences count. **In:** s ; p

Out: number of occurrences

Constraints: $1 \leq |s|, |p| \leq 10^6$.

1732 — Finding Borders

A border is a proper prefix that is also a suffix. Find the lengths of all borders of a lowercase string. **In:** s

Out: all border lengths in increasing order

Constraints: $1 \leq |s| \leq 10^6$.

1733 — Finding Periods

A period is a prefix whose repetitions generate the whole string; the final repetition may be partial. Find all valid period lengths (including $|s|$). **In:** s

Out: all period lengths in increasing order

Constraints: $1 \leq |s| \leq 10^6$.

1110 — Minimal Rotation

Among all cyclic rotations of a lowercase string, output the lexicographically smallest rotation. **In:** s

Out: lexicographically minimal rotation

Constraints: $1 \leq |s| \leq 10^6$.

1111 — Longest Palindrome

Find a longest palindromic *substring* of a lowercase string. If several exist, output any one. **In:** s

Out: any longest palindromic substring

Constraints: $1 \leq |s| \leq 10^6$.

3138 — All Palindromes

For every position i , find the maximum length of a palindromic substring that ends exactly at i . **In:** s

Out: n lengths, for positions $1, \dots, n$

Constraints: $1 \leq n \leq 2 \cdot 10^5$.

1112 — Required Substring

Count uppercase A–Z strings of length n that contain a given pattern p as a contiguous substring at least once. **In:** n ; pattern p

Out: count mod $10^9 + 7$

Constraints: $1 \leq n \leq 1000$, $1 \leq |p| \leq 100$.

2420 — Palindrome Queries

Maintain a lowercase string under: 1 **k** **x** set $s_k = x$; 2 **a** **b** ask whether $s[a..b]$ is a palindrome. **In:** n, m ; s ; then m operations

Out: YES/NO for each type 2 query

Constraints: $n, m \leq 2 \cdot 10^5$.

2102 — Finding Patterns

Given a lowercase text and many lowercase patterns, decide independently whether each pattern occurs as a substring. **In:** s ; k ; then k patterns

Out: YES or NO for each pattern

Constraints: $|s| \leq 10^5$, $k \leq 5 \cdot 10^5$; total pattern length $\leq 5 \cdot 10^5$.

2103 — Counting Patterns

For each of many patterns, count its starting positions in a fixed text; overlapping occurrences count. **In:** s ; k ; then k patterns

Out: occurrence count for each pattern

Constraints: $|s| \leq 10^5$, $k \leq 5 \cdot 10^5$; total pattern length $\leq 5 \cdot 10^5$.

2104 — Pattern Positions

For each pattern, find the first 1-indexed position where it occurs in a fixed text, or -1 if absent. **In:** s ; k ; then k patterns

Out: first position or -1 for each pattern

Constraints: $|s| \leq 10^5$, $k \leq 5 \cdot 10^5$; total pattern length $\leq 5 \cdot 10^5$.

2105 — Distinct Substrings

Count the distinct nonempty substrings of a lowercase string. **In:** s

Out: number of distinct substrings

Constraints: $1 \leq |s| \leq 10^5$; answer fits 64-bit.

1149 — Distinct Subsequences

Delete arbitrary characters without reordering the rest. Count the distinct *nonempty* strings obtainable as subsequences. **In:** s

Out: number of distinct nonempty subsequences mod $10^9 + 7$

Constraints: $1 \leq |s| \leq 5 \cdot 10^5$.

2106 — Repeating Substring

Find a longest substring that occurs in at least two positions of s (occurrences may overlap). If none exists, print -1 . **In:** s

Out: any longest repeating substring, or -1

Constraints: $1 \leq |s| \leq 10^5$.

2107 — String Functions

For every 1-indexed position i , compute $z(i)$ = longest prefix match starting at i (with $z(1) = 0$), and $\pi(i)$ = longest proper prefix ending at i . **In:** s

Out: one line of all z values; one line of all π values

Constraints: $1 \leq |s| \leq 10^6$.

3225 — Inverse Suffix Array

Given a permutation of $1, \dots, n$ claimed to be the lexicographic order of suffix starting positions, construct any lowercase a–z string having exactly that suffix array. **In:** n ; suffix-array permutation

Out: any valid lowercase string; -1 if impossible

Constraints: $1 \leq n \leq 10^5$.

1113 — String Transform

Invert the Burrows–Wheeler transform defined by appending unique sentinel $\#$ (smaller than a–z), sorting all rotations, then taking their last characters. **In:** transformed string of length $n + 1$, containing one $\#$

Out: original lowercase string of length n

Constraints: $1 \leq n \leq 10^6$.

2108 — Substring Order I

Sort all *distinct* nonempty substrings lexicographically and output the k th one. **In:** s ; k

Out: k th distinct substring

Constraints: $|s| \leq 10^5$, $1 \leq k \leq n(n + 1)/2$; k is guaranteed valid.

2109 — Substring Order II

Sort all substring *occurrences* lexicographically (duplicates appear with multiplicity) and output the k th one. **In:** s ; k

Out: k th substring in the multiset order

Constraints: $|s| \leq 10^5$, $1 \leq k \leq n(n + 1)/2$.

2110 — Substring Distribution

For each length $L = 1, \dots, n$, count how many distinct substrings of s have length exactly L . **In:** s

Out: n counts for lengths $1, \dots, n$

Constraints: $1 \leq |s| \leq 10^5$.

CSES — Geometry Problem Reference

All tasks: 1 s, 512 MiB. Coordinates and answers are exact integers unless stated otherwise.

1740 — Intersection Points

Given n horizontal/vertical segments, count their intersection points. Parallel segments never intersect, and no segment endpoint is an intersection point. **In:** n ; then endpoints $(x_1, y_1), (x_2, y_2)$

Out: number of intersections

Constraints: $n \leq 10^5$; coordinates in $[-10^6, 10^6]$; each segment is nonempty.

1741 — Area of Rectangles

Given n axis-aligned rectangles, find the total area of their union; overlapping area is counted once. **In:** n ; then opposite corners $(x_1, y_1), (x_2, y_2)$

Out: union area

Constraints: $n \leq 10^5$; $-10^6 \leq x_1 < x_2 \leq 10^6$, same for y .

1742 — Robot Path

A robot starts at $(0, 0)$ and follows axis-aligned commands. It stops after all commands or immediately upon reaching any point visited earlier, possibly partway through a command. Find distance traveled until stopping. **In:** n ; then direction $d \in \{U, D, L, R\}$ and distance x

Out: total distance traveled

Constraints: $n \leq 10^5$, $1 \leq x \leq 10^6$.

2189 — Point Location Test

For directed line $p_1 \rightarrow p_2$, classify point p_3 as lying to its left, right, or on the line. **In:** t ; then p_1, p_2, p_3 coordinates

Out: LEFT, RIGHT, or TOUCH per test

Constraints: $t \leq 10^5$; coordinates in $[-10^9, 10^9]$; $p_1 \neq p_2$.

2190 — Line Segment Intersection

For two closed nondegenerate segments, determine whether they share at least one point. Endpoint touching and collinear overlap both count as intersection. **In:** t ; then endpoints of both segments

Out: YES/NO per test

Constraints: $t \leq 10^5$; coordinates in $[-10^9, 10^9]$.

2191 — Polygon Area

Given the vertices of a simple polygon in boundary order, compute its area exactly. **In:** n ; then vertices (x_i, y_i)

Out: $2A$, where A is the polygon area

Constraints: $3 \leq n \leq 1000$; coordinates in $[-10^9, 10^9]$.

2192 — Point in Polygon

Given a simple polygon and m query points, classify each point as inside, outside, or on the polygon boundary. **In:** n, m ; n polygon vertices; then m points

Out: INSIDE, OUTSIDE, or BOUNDARY per point

Constraints: $n, m \leq 1000$; coordinates in $[-10^9, 10^9]$.

2193 — Polygon Lattice Points

For a simple polygon with integer-coordinate vertices, count lattice points strictly inside it and lattice points on its boundary.

In: n ; then integer vertices (x_i, y_i)

Out: two integers: interior count, boundary count

Constraints: $3 \leq n \leq 10^5$; coordinates in $[-10^9, 10^9]$.

2194 — Minimum Euclidean Distance

Among n distinct planar points, find the minimum Euclidean distance between two different points. **In:** n ; then points (x, y)

Out: d^2 for the minimum distance d

Constraints: $2 \leq n \leq 2 \cdot 10^5$; coordinates in $[-10^9, 10^9]$.

2195 — Convex Hull

Find the convex hull of n distinct points. Output *all* input points lying on the hull boundary, including collinear points along hull edges; any output order is accepted. **In:** n ; then points (x, y)

Out: k , then the k hull-boundary points

Constraints: $3 \leq n \leq 2 \cdot 10^5$; hull area is positive; coordinates in $[-10^9, 10^9]$.

3410 — Maximum Manhattan Distances

Points are inserted one by one into an initially empty set. After each insertion, report the maximum Manhattan distance $|x_1 - x_2| + |y_1 - y_2|$ between any two inserted points. **In:** n ; then the points in insertion order

Out: one maximum distance after each insertion

Constraints: $n \leq 2 \cdot 10^5$; points distinct; coordinates in $[-10^9, 10^9]$.

3411 — All Manhattan Distances

Given n distinct points, compute the sum of Manhattan distances over all unordered pairs. **In:** n ; then points (x, y)

Out: $\sum_{i < j} (|x_i - x_j| + |y_i - y_j|)$

Constraints: $n \leq 2 \cdot 10^5$; coordinates in $[-10^9, 10^9]$.

3427 — Line Segments Trace I

Each segment runs from $(0, y_1)$ to (m, y_2) and has integer slope. For every integer $x = 0, 1, \dots, m$, find the maximum y attained by any segment there. **In:** n, m ; then y_1, y_2 for each segment

Out: $m + 1$ maxima for $x = 0, \dots, m$

Constraints: $n, m \leq 10^5$, $0 \leq y_1, y_2 \leq 10^9$.

3428 — Line Segments Trace II

Segments have integer-coordinate endpoints, integer slope, and arbitrary active intervals $[x_1, x_2] \subseteq [0, m]$. For every integer x , find the maximum y among segments covering that x ; output -1 if none does. **In:** n, m ; then $(x_1, y_1), (x_2, y_2)$

Out: $m + 1$ values for $x = 0, \dots, m$

Constraints: $n, m \leq 10^5$; $0 \leq x_1 < x_2 \leq m$; $0 \leq y_1, y_2 \leq 10^9$.

3429 — Lines and Queries I

Process online queries: 1 **a b** adds the infinite line $y = ax + b$; 2 **x** asks the maximum value among all lines at that x . The first query is always an insertion. **In:** n ; then n queries

Out: maximum value for each type 2 query

Constraints: $n \leq 2 \cdot 10^5$; $|a|, |b| \leq 10^9$; $0 \leq x \leq 10^5$.

3430 — Lines and Queries II

Process online queries: 1 **a b l r** adds $y = ax + b$ active only for $x \in [l, r]$; 2 **x** asks the maximum among lines active at x . **In:** n ; then n queries

Out: maximum for each type 2 query, or NO if no line is active

Constraints: $n \leq 2 \cdot 10^5$; $|a|, |b| \leq 10^9$; $0 \leq x, l, r \leq 10^5$, $l \leq r$.

CSES – Advanced Techniques Problem Reference

All tasks: 1 s, 512 MB. Counts are modulo $10^9 + 7$ only where explicitly stated.

1628 – Meet in the Middle

Given n positive integers, count subsets of the array elements whose sum is exactly x ; equal values at different positions are distinct choices. **In:** n x ; $t_1 \dots t_n$

Out: number of subsets with sum x

Constraints: $n \leq 40$; $x, t_i \leq 10^9$. Use 64-bit count.

2136 – Hamming Distance

Given n binary strings of equal length k , find the minimum Hamming distance between any two distinct strings. **In:** n k ; then n bit strings

Out: minimum number of differing positions

Constraints: $n \leq 2 \cdot 10^4$; $k \leq 30$.

3360 – Corner Subgrid Check

An $n \times n$ grid uses the first k uppercase letters. For each letter, determine whether some subgrid of height and width at least 2 has that letter in all four corners. **In:** n k ; then n grid rows

Out: k lines: YES/NO for each letter

Constraints: $n \leq 3000$; $k \leq 26$.

2137 – Corner Subgrid Count

In an $n \times n$ binary grid, count subgrids of height and width at least 2 whose four corner cells are all black (1). **In:** n ; then n binary rows

Out: number of beautiful subgrids

Constraints: $n \leq 3000$. Use 64-bit count.

2138 – Reachable Nodes

Given a directed acyclic graph, for every node count how many nodes are reachable from it, including the node itself.

In: n m ; then m directed edges a b

Out: n reachability counts

Constraints: $n \leq 5 \cdot 10^4$; $m \leq 10^5$.

2143 – Reachability Queries

Given a directed graph, answer whether node b is reachable from node a for each query. **In:** n m q ; m directed edges; then q pairs a b

Out: YES or NO per query

Constraints: $n \leq 5 \cdot 10^4$; $m, q \leq 10^5$.

2072 – Cut and Paste

Maintain a string under operations that cut the current substring at positions $[a, b]$, remove it, and append it to the end. Output the final string. **In:** n m ; uppercase string; then m lines a b

Out: final string

Constraints: $n, m \leq 2 \cdot 10^5$.

2073 – Substring Reversals

Maintain a string under operations that reverse the current substring at positions $[a, b]$. Output the final string. **In:** n m ; uppercase string; then m lines a b

Out: final string

Constraints: $n, m \leq 2 \cdot 10^5$.

2074 – Reversals and Sums

Maintain an integer array with operations: 1 a b reverses subarray $[a, b]$; 2 a b asks its sum. **In:** n m ; array; then m operations

Out: sum for every type-2 operation

Constraints: $n \leq 2 \cdot 10^5$; $m \leq 10^5$; $0 \leq x_i \leq 10^9$. Use 64-bit sums.

2076 – Necessary Roads

A connected simple undirected graph is given. Find every road whose removal makes some pair of cities disconnected.

In: n m ; then m undirected edges

Out: number of necessary roads, then the roads in any order

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

2077 – Necessary Cities

A connected simple undirected graph is given. Find every city whose removal, together with its incident roads, leaves some two other cities disconnected. **In:** n m ; then m undirected edges

Out: number of necessary cities, then their list in any order

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

2078 – Eulerian Subgraphs

Choose any subset of edges of an undirected graph while keeping all vertices. Count choices for which every vertex has even degree. **In:** n m ; then m undirected edges

Out: number of Eulerian subgraphs modulo $10^9 + 7$

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

2084 – Monster Game I

Process levels in order. At level $i < n$, kill or escape; level n must be killed. Killing costs $s_i \cdot f$ using the current skill factor f , then sets $f = f_i$; escaping changes nothing. Minimize total time. **In:** n x ; strengths s_i ; resulting skill factors f_i

Out: minimum total time

Constraints: $n \leq 2 \cdot 10^5$; $s_i \leq 10^6$ nondecreasing; $x \geq f_1 \geq \dots \geq f_n \geq 1$. Use 64-bit.

CSES – Advanced Techniques Problem Reference

Continued. All tasks: 1 s, 512 MB.

2085 – Monster Game II

Same game as Monster Game I: kill/escape levels in order, final monster mandatory; killing i costs s_i times the current skill factor and then sets it to f_i . Here strengths and skill factors have no monotonicity guarantees. **In:** $\mathbf{n\ x}$; strengths s_i ; resulting factors f_i

Out: minimum total time

Constraints: $n \leq 2 \cdot 10^5$; $x, s_i, f_i \leq 10^6$. Use 64-bit.

2086 – Subarray Squares

Partition an array into exactly k nonempty contiguous subarrays. A part costs the square of the sum of its elements; minimize the total cost. **In:** $\mathbf{n\ k; x_1 \dots x_n}$

Out: minimum total cost

Constraints: $1 \leq k \leq n \leq 3000$; $x_i \leq 10^5$. Use 64-bit.

2087 – Houses and Schools

Houses are at positions $1 \dots n$ with c_i children at house i . Place k schools in houses; every child walks to a nearest school. Minimize total walking distance. **In:** $\mathbf{n\ k; c_1 \dots c_n}$

Out: minimum total distance

Constraints: $1 \leq k \leq n \leq 3000$; $c_i \leq 10^9$. Use 64-bit.

2088 – Knuth Division

Starting with the whole array as one subarray, repeatedly split one current subarray into two until all parts are singletons. Each split costs the sum of the values in the subarray being split. Minimize total cost. **In:** $\mathbf{n; x_1 \dots x_n}$

Out: minimum total split cost

Constraints: $n \leq 5000$; $x_i \leq 10^9$. Use 64-bit.

2111 – Apples and Bananas

There are n apples and m bananas, each with integer weight $1 \dots k$. For every total weight $w = 2 \dots 2k$, count pairs consisting of one apple and one banana with combined weight w . **In:** $\mathbf{k\ n\ m}$; apple weights; banana weights

Out: $2k - 1$ counts for totals $2 \dots 2k$

Constraints: $k, n, m \leq 2 \cdot 10^5$. Use 64-bit counts.

2112 – One Bit Positions

Given a binary string, for every distance $d = 1 \dots n - 1$, count pairs of positions i, j with $i - j = d$ and both characters equal to 1. **In:** one binary string of length n

Out: $n - 1$ counts, for distances $1 \dots n - 1$

Constraints: $2 \leq n \leq 2 \cdot 10^5$.

2113 – Signal Processing

Slide a mask across a signal from left to right, including partial overlaps at both ends. At every alignment, output the sum of products of all aligned signal/mask values. **In:** $\mathbf{n\ m}$; signal $a_1 \dots a_n$; mask $b_1 \dots b_m$

Out: $n + m - 1$ overlap dot products in order

Constraints: $n, m \leq 2 \cdot 10^5$; $a_i, b_i \leq 100$. Use 64-bit sums.

2101 – New Roads Queries

Initially there are no roads; one undirected road is added each day in the given order. For each pair (a, b) , find the first day when a and b become connected. **In:** $\mathbf{n\ m\ q}$; m roads in build order; then q pairs

Out: earliest day for each query, or -1 if never connected

Constraints: $n, m, q \leq 2 \cdot 10^5$.

2133 – Dynamic Connectivity

Maintain an undirected simple graph under edge insertions and deletions. Report the number of connected components before any event and after every event. **In:** $\mathbf{n\ m\ k}$; m initial edges; then k events $\mathbf{t\ a\ b}$

Out: $k + 1$ component counts

Constraints: $n \leq 10^5$; $m, k \leq 10^5$; updates are always valid.

2121 – Parcel Delivery

A directed route $a \rightarrow b$ has capacity r parcels and cost c per parcel. Send exactly k parcels from city 1 to city n at minimum total cost. **In:** $\mathbf{n\ m\ k}$; then routes $\mathbf{a\ b\ r\ c}$

Out: minimum cost, or -1 if sending k parcels is impossible

Constraints: $n \leq 500$; $m \leq 1000$; $k \leq 100$; $r, c \leq 1000$.

2129 – Task Assignment

There are n employees and n tasks with cost c_{ij} for assigning task j to employee i . Assign every employee exactly one distinct task, minimizing total cost, and output one optimal assignment. **In:** $\mathbf{n}$; then $n \times n$ cost matrix

Out: minimum cost; then n employee–task pairs

Constraints: $n \leq 200$; $1 \leq c_{ij} \leq 1000$.

2130 – Distinct Routes II

A directed graph has teleporters that may each be used at most once over the entire game. Play exactly k days; each day starts at node 1 and must reach node n . Every teleporter use costs one coin. Minimize total coins and output the k routes. **In:** $\mathbf{n\ m\ k}$; then m directed teleporters $\mathbf{a\ b}$

Out: minimum total uses and k valid routes, or only -1 if impossible

Constraints: $n \leq 500$; $m \leq 1000$; $1 \leq k \leq n - 1$.

CSES Sliding Window Problems — Problem Reference

All tasks: 1 s, 512 MiB. Windows are processed from left to right.

1076 — Sliding Window Median

For every length- k window of $x_1, \dots, x_n$, find its median. If k is even, use the smaller of the two middle elements. **In:** n k ; $x_1 \dots x_n$

Out: $n - k + 1$ medians

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.

1077 — Sliding Window Cost

For every length- k window, find the minimum total cost to make all elements equal, where changing u to v costs $|u - v|$.

In: n k ; $x_1 \dots x_n$

Out: $n - k + 1$ minimum costs

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

3219 — Sliding Window Mex

For every length- k window, find its mex: the smallest nonnegative integer not present in the window. **In:** n k ; $x_1 \dots x_n$

Out: $n - k + 1$ mex values

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $0 \leq x_i \leq 10^9$.

3220 — Sliding Window Sum

The array is generated online. For every length- k window compute its sum, then xor all window sums together. **In:** n k ; x a b c

Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \leq i \leq n$.

Out: xor of all window sums

Constraints: $1 \leq k \leq n \leq 10^7$, $0 \leq x, a, b \leq 10^9$, $1 \leq c \leq 10^9$. Use 64-bit sums.

3221 — Sliding Window Minimum

The array is generated online. For every length- k window compute its minimum, then xor all window minimums together. **In:** n k ; x a b c

Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \leq i \leq n$.

Out: xor of all window minimums

Constraints: $1 \leq k \leq n \leq 10^7$, $0 \leq x, a, b \leq 10^9$, $1 \leq c \leq 10^9$.

3222 — Sliding Window Distinct Values

For every length- k window, count how many distinct values it contains. **In:** n k ; $x_1 \dots x_n$

Out: $n - k + 1$ distinct-value counts

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.

3223 — Sliding Window Inversions

For every length- k window, count inversions: pairs of positions $p < q$ inside the window with $x_p > x_q$. **In:** n k ; $x_1 \dots x_n$

Out: $n - k + 1$ inversion counts

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

3224 — Sliding Window Mode

For every length- k window, output its most frequent value. If several values have maximum frequency, choose the smallest.

In: n k ; $x_1 \dots x_n$

Out: $n - k + 1$ modes

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.

3227 — Sliding Window Advertisement

A fence has n unit-width boards of heights x_i . For every k consecutive boards, find the largest axis-aligned rectangle that fits under those bars. **In:** n k ; $x_1 \dots x_n$

Out: $n - k + 1$ maximum rectangle areas

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$. Use 64-bit.

3405 — Sliding Window Or

The array is generated online. For every length- k window compute the bitwise OR of its elements, then xor all window ORs together. **In:** n k ; x a b c

Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \leq i \leq n$.

Out: xor of all window ORs

Constraints: $1 \leq k \leq n \leq 10^7$, $0 \leq x, a, b \leq 10^9$, $1 \leq c \leq 10^9$.

3426 — Sliding Window Xor

The array is generated online. For every length- k window compute the bitwise XOR of its elements, then xor all window XORs together. **In:** n k ; x a b c

Generator: $x_1 = x$, $x_i = (ax_{i-1} + b) \bmod c$ for $2 \leq i \leq n$.

Out: xor of all window XORs

Constraints: $1 \leq k \leq n \leq 10^7$, $0 \leq x, a, b \leq 10^9$, $1 \leq c \leq 10^9$.

CSES – Interactive Problems Reference

All tasks: 1 s, 512 MB. Flush after every output line; respect the query budget and terminate immediately when finished.

3112 – Hidden Integer

A hidden integer x is fixed. Determine x by asking threshold questions of the form “is $y < x$?”. **Start:** no initial input; begin querying immediately

Ask: print ? y , $1 \leq y \leq 10^9$

Reply: YES iff $y < x$, otherwise NO

Finish: print ! x and terminate

Constraints: $1 \leq x \leq 10^9$; at most 30 ? queries.

3139 – Hidden Permutation

A hidden permutation $a_1, \dots, a_n$ of $1, \dots, n$ is fixed. Recover every value by comparing entries at chosen indices. **Start:** read n

Ask: print ? i j to ask whether $a_i < a_j$

Reply: YES iff $a_i < a_j$, otherwise NO

Finish: print ! $a_1 \dots a_n$ and terminate

Constraints: $n \leq 1000$; at most 10^4 comparison queries.

3305 – K-th Highest Score

There are n distinct Finnish scores and n distinct Swedish scores. Within each country, query scores by descending rank; find the k -th highest score among all $2n$ coders. **Start:** read n, k

Ask: print F i or S i for that country’s i -th highest score

Reply: the requested integer score

Finish: print ! s , where s is the overall k -th highest score, then terminate

Constraints: $n \leq 10^5$, $1 \leq k \leq 2n$; all $2n$ scores distinct, in $[1, 10^9]$; at most 100 score queries total.

3228 – Permuted Binary Strings

A hidden permutation $a_1, \dots, a_n$ is fixed. For any chosen binary string $b_1 \dots b_n$, the grader returns the string permuted as $b_{a_1} \dots b_{a_n}$. Recover the permutation. **Start:** read n

Ask: print ? b , where $b \in \{0, 1\}^n$

Reply: the length- n string $b_{a_1} \dots b_{a_n}$

Finish: print ! $a_1 \dots a_n$ and terminate

Constraints: $n \leq 1000$; at most 10 binary-string queries.

3273 – Colored Chairs

n chairs form a cycle; every chair is red or blue, and n is odd. Find an index i such that adjacent chairs i and $i + 1$ have the same color, with $n + 1 \equiv 1$. **Start:** read n

Ask: print ? i to query chair i

Reply: R or B

Finish: print ! i asserting chairs i and $i + 1$ have equal color, then terminate

Constraints: $3 \leq n \leq 2 \cdot 10^5$, n odd; at most 20 color queries.

3140 – Inversion Sorting

A hidden permutation of $1, \dots, n$ must be sorted. Each move reverses a chosen subarray; afterward the grader reports the permutation’s new inversion count. You win as soon as it becomes 0. **Start:** read n

Ask: print i j to reverse positions $i \dots j$

Reply: the inversion count after that reversal

Finish: when the returned count is 0, terminate immediately; there is no separate ! command

Constraints: $n \leq 1000$; at most $4n$ reversal operations.

CSES — Bitwise Operations Problem Reference

All tasks: 1 s, 512 MiB.

1146 — Counting Bits

For all integers $1 \leq v \leq n$, count the total number of set bits in their binary representations.

In: n

Out: total number of 1 bits over $1, \dots, n$

Constraints: $1 \leq n \leq 10^{15}$. Use 64-bit.

1655 — Maximum Xor Subarray

Given $x_1, \dots, x_n$, find the maximum bitwise XOR of any nonempty contiguous subarray.

In: n ; $x_1 \dots x_n$

Out: maximum subarray XOR

Constraints: $n \leq 2 \cdot 10^5$, $0 \leq x_i \leq 10^9$.

3191 — Maximum Xor Subset

Choose any subset of the array and XOR its elements. Find the maximum obtainable XOR value.

In: n ; $x_1 \dots x_n$

Out: maximum subset XOR

Constraints: $n \leq 2 \cdot 10^5$, $0 \leq x_i \leq 10^9$.

3211 — Number of Subset Xors

Consider XOR values of all subsets, including the empty subset with XOR 0. Count how many distinct values are obtainable.

In: n ; $x_1 \dots x_n$

Out: number of distinct subset XORs

Constraints: $n \leq 2 \cdot 10^5$, $0 \leq x_i \leq 10^9$.

3192 — K Subset Xors

Consider the XOR of every one of the 2^n subsets, including the empty subset. Values are counted with multiplicity. Print the k smallest values in nondecreasing order.

In: n k ; $x_1 \dots x_n$

Out: k smallest subset XORs, including duplicates

Constraints: $n \leq 2 \cdot 10^5$, $k \leq \min(2^n, 2 \cdot 10^5)$, $0 \leq x_i \leq 10^9$.

3233 — All Subarray Xors

Find every distinct integer that occurs as the XOR of some nonempty contiguous subarray.

In: n ; $x_1 \dots x_n$

Out: first count k , then the k XOR values in increasing order

Constraints: $n \leq 2 \cdot 10^5$, $0 \leq x_i \leq 10^6$.

2419 — Xor Pyramid Peak

The bottom row is $a_1, \dots, a_n$. Every cell above is XOR of its lower-left and lower-right cells. Find the single top cell.

In: n ; $a_1 \dots a_n$

Out: topmost value

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq a_i \leq 10^9$.

3194 — Xor Pyramid Diagonal

Build the same XOR pyramid. Print the *leftmost* value of every row, ordered from the bottom row up to the top.

In: n ; $a_1 \dots a_n$

Out: n leftmost row values, bottom $\rightarrow$ top

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq a_i \leq 10^9$.

3195 — Xor Pyramid Row

Build the XOR pyramid from the given bottom row. Print all values on the k th row *from the top*; that row has exactly k entries.

In: n k ; $a_1 \dots a_n$

Out: the k values on row k from the top

Constraints: $1 \leq k \leq n \leq 2 \cdot 10^5$, $1 \leq a_i \leq 10^9$.

1654 — SOS Bit Problem

For every input element x , count array elements y satisfying each condition:

- $x \mid y = x$;
- $x \& y = x$;
- $x \& y \neq 0$.

Counts include occurrences, so equal array values are counted separately.

In: n ; $x_1 \dots x_n$

Out: three counts for each input element, in input order

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^6$.

3141 — And Subset Count

For every $k = 0, 1, \dots, n$, count the nonempty subsets whose bitwise AND of all chosen elements is exactly k .

In: n ; $a_1 \dots a_n$

Out: $n + 1$ counts for $k = 0, \dots, n$, modulo $10^9 + 7$

Constraints: $n \leq 2 \cdot 10^5$, $0 \leq a_i \leq n$.

CSES — Construction Problems Reference

All tasks: 1 s, 512 MiB. Any valid construction is accepted unless stated otherwise.

2214 — Inverse Inversions

Construct a permutation p of $1, \dots, n$ having exactly k inversions, where an inversion is a pair $i < j$ with $p_i > p_j$.

In: n k

Out: one valid permutation

Constraints: $1 \leq n \leq 10^6$, $0 \leq k \leq n(n-1)/2$.

2215 — Monotone Subsequences

For each test, construct a permutation of $1, \dots, n$ whose *longest monotone subsequence* has exactly k elements. A monotone subsequence may be increasing or decreasing.

In: t ; then n k

Out: a valid permutation for each test, or IMPOSSIBLE

Constraints: $t \leq 1000$, $1 \leq k \leq n \leq 100$.

3422 — Third Permutation

Given permutations a, b with $a_i \neq b_i$ for every position, construct a permutation c such that simultaneously $c_i \neq a_i$ and $c_i \neq b_i$ for every i .

In: n ; permutation a ; permutation b

Out: a valid permutation c , or IMPOSSIBLE

Constraints: $2 \leq n \leq 10^5$.

3423 — Permutation Prime Sums

Construct two permutations a, b of $1, \dots, n$ such that $a_i + b_i$ is prime for every position i .

In: n

Out: the two permutations, or IMPOSSIBLE

Constraints: $1 \leq n \leq 10^5$.

1697 — Chess Tournament

Player i wants to play exactly x_i games. Choose unordered pairs of distinct players so every player gets the requested degree; each pair may play at most once.

In: n ; $x_1 \dots x_n$

Out: number k of games and the k player pairs; or IMPOSSIBLE

Constraints: $n \leq 10^5$, $\sum_i x_i \leq 2 \cdot 10^5$.

3424 — Distinct Sums Grid

Construct an $n \times n$ integer grid such that:

- every value $1, \dots, n$ appears exactly n times overall;
- the n row sums and n column sums are all pairwise distinct, giving exactly $2n$ distinct sums.

In: n

Out: a valid grid, or IMPOSSIBLE

Constraints: $1 \leq n \leq 1000$.

2423 — Filling Trominos

Tile the entire $n \times m$ grid with L-trominos, each occupying exactly three cells in an L shape. If possible, output letters so two edge-adjacent cells have the same letter *iff* they belong to the same tromino.

In: t ; then n m

Out: for each test, YES plus an n -row letter grid, or NO

Constraints: $t \leq 100$, $1 \leq n, m \leq 100$.

2418 — Grid Path Construction

Given distinct squares $a = (y_1, x_1)$ and $b = (y_2, x_2)$ in an $n \times m$ grid, construct a path from a to b that visits *every grid square exactly once*. Moves are only U/D/L/R.

In: t ; then n m y_1 x_1 y_2 x_2

Out: for each test, YES and a move string of length $nm - 1$, or NO

Constraints: $t \leq 100$, $1 \leq n, m \leq 50$; endpoints are distinct.

CSES – Advanced Graph Problems Reference

All tasks: 1 s, 512 MB. Paths/walks and output witnesses are stated explicitly where relevant.

3303 – Nearest Shops

An undirected graph has anime shops in k cities. For every city, find the shortest distance to a shop in a *different* city; a local shop does not count for that city's answer. **In:** n m k ; k shop cities; then m undirected edges

Out: n distances; -1 if no other shop is reachable

Constraints: $k \leq n \leq 10^5$; $m \leq 2 \cdot 10^5$. Graph may be disconnected.

1134 – Prüfer Code

Decode a Prüfer code of length $n - 2$ into its labeled tree. The code convention repeatedly removes the smallest-labeled current leaf and records its neighbor. **In:** n ; then $n - 2$ code values

Out: $n - 1$ tree edges in any order

Constraints: $3 \leq n \leq 2 \cdot 10^5$; code values in $[1, n]$.

1702 – Tree Traversals

A binary tree has distinct labels. Given its preorder and inorder traversals, reconstruct and print the postorder traversal. **In:** n ; preorder; inorder

Out: postorder traversal

Constraints: $n \leq 10^5$; input is guaranteed to describe a valid binary tree.

1757 – Course Schedule II

Given prerequisite edges $a \rightarrow b$, output the valid course order that completes course 1 as early as possible, then course 2 as early as possible, and so on. This priority is by course number, not simply by the next available label. **In:** n m ; then m prerequisites a b

Out: n courses in the required order

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$; at least one valid schedule exists.

1756 – Acyclic Graph Edges

Orient every edge of an undirected graph so that the resulting directed graph has no directed cycle. **In:** n m ; then m undirected edges

Out: m directed edges; any valid orientation

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

2177 – Strongly Connected Edges

Orient every edge of a simple undirected graph so that the resulting directed graph is strongly connected. **In:** n m ; then m undirected edges

Out: a valid orientation, or IMPOSSIBLE

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

2179 – Even Outdegree Edges

Orient every edge of a simple undirected graph so that every vertex has even outdegree. **In:** n m ; then m undirected edges

Out: a valid orientation, or IMPOSSIBLE

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

1707 – Graph Girth

Find the length (number of edges) of the shortest cycle in a simple undirected graph. **In:** n m ; then m undirected edges

Out: girth, or -1 if the graph is acyclic

Constraints: $n \leq 2500$; $m \leq 5000$.

3357 – Fixed Length Walk Queries

For each query (a, b, x) in a connected simple undirected graph, determine whether a *walk* of exactly x edges can start at a and end at b ; vertices/edges may be revisited. **In:** n m q ; edges; then queries a b x

Out: YES/NO per query

Constraints: $n \leq 2500$; $m \leq 5000$; $q \leq 10^5$; $0 \leq x \leq 10^9$.

3111 – Transfer Speeds Sum

A weighted tree has edge transfer speeds. For vertices a, b , $d(a, b)$ is the minimum edge speed on their unique path. Sum $d(a, b)$ over all unordered vertex pairs. **In:** n ; then $n - 1$ edges a b x

Out: $\sum_{a < b} d(a, b)$

Constraints: $n \leq 2 \cdot 10^5$; $x \leq 10^6$. Use 64-bit.

3407 – MST Edge Check

For every edge of a connected weighted simple graph, determine whether it can belong to *some* minimum spanning tree. **In:** n m ; then edges a b w

Out: YES/NO for each edge in input order

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$; $w \leq 10^9$.

3408 – MST Edge Set Check

For each queried set of edge indices, determine whether there exists one MST that contains *all* edges in that set simultaneously. **In:** n m q ; weighted edges; then q edge-index sets

Out: YES/NO per set

Constraints: $n \leq 10^5$; $m, q \leq 2 \cdot 10^5$; total queried-set size $\leq m$.

3409 – MST Edge Cost

For every edge, find the minimum possible spanning-tree cost under the condition that this edge must be included. **In:** n m ; then edges a b w

Out: conditioned minimum cost for each edge in input order

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$; $w \leq 10^9$. Use 64-bit.

1677 – Network Breakdown

An undirected network starts with m edges; k specified connections break one by one. Report the number of connected components after each deletion. **In:** n m k ; initial edges; then k broken edges

Out: k component counts, in breakdown order

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$; $1 \leq k \leq m$.

3114 – Tree Coin Collecting I

A tree has some coin vertices. For each (a, b) , find the minimum length of a walk from a to b that visits *at least one* coin vertex; detours are allowed. **In:** n q ; coin flags; tree edges; then q pairs a b

Out: minimum walk length per query

Constraints: $n, q \leq 2 \cdot 10^5$; at least one coin exists.

3149 – Tree Coin Collecting II

A tree has some coin vertices. For each (a, b) , find the minimum length of a walk from a to b that visits *every* coin vertex. **In:** n q ; coin flags; tree edges; then q pairs a b

Out: minimum walk length per query

Constraints: $n, q \leq 2 \cdot 10^5$; at least one coin exists. Use 64-bit if desired.

1700 – Tree Isomorphism I

For each test, compare two rooted trees of equal size. Node 1 is the root in both; determine whether the rooted tree shapes are isomorphic (labels otherwise irrelevant). **In:** t ; per test: n , then edges of tree 1 and tree 2

Out: YES/NO per test

Constraints: sum of $n \leq 10^5$; $2 \leq n \leq 10^5$.

1701 – Tree Isomorphism II

For each test, compare two *unrooted* trees of equal size and determine whether their shapes are isomorphic; vertex labels are irrelevant. **In:** t ; per test: n , then edges of tree 1 and tree 2

Out: YES/NO per test

Constraints: sum of $n \leq 10^5$; $2 \leq n \leq 10^5$.

1699 – Flight Route Requests

Initially there are no directed flights. Given reachability requests $a \rightsquigarrow b$, find the minimum number of one-way flight connections whose transitive reachability satisfies every request. **In:** n m ; then m distinct requested pairs a b

Out: minimum number of flights needed

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

1703 – Critical Cities

In a directed graph with a route from 1 to n , find all cities that occur on *every* route from 1 to n . **In:** n m ; then m directed edges

Out: count, then critical cities in increasing order

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

1203 – Visiting Cities

In a directed positive-weight graph, consider only minimum-price routes from city 1 to city n . Find all cities that occur on *every shortest-price route*. **In:** n m ; then flights a b c

Out: count, then mandatory cities in increasing order

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$; $c \leq 10^9$; a route $1 \rightarrow n$ exists.

3308 – Graph Coloring

Color a simple graph using the minimum possible number of colors so adjacent vertices differ, and output an optimal coloring. **In:** n m ; then m undirected edges

Out: chromatic number k , then n colors in $[1, k]$

Constraints: $n \leq 16$; $m \leq n(n-1)/2$.

3158 – Bus Companies

Company j sells a ticket of cost c_j that lets you travel directly between any two cities it serves. Starting from city 1, find the cheapest route cost to every city. **In:** n m ; costs c_j ; for each company, its number/list of served cities

Out: n minimum costs from city 1

Constraints: $n, m \leq 10^5$; total company-city incidences $\leq 2 \cdot 10^5$; $c_j \leq 10^9$.

3358 – Split into Two Paths

Given a DAG, partition all vertices into exactly two vertex-disjoint directed paths so every vertex appears in exactly one path. One path may be empty; unused graph edges do not matter. **In:** n m ; then m directed edges

Out: YES and the two path vertex lists, or NO

Constraints: $n \leq 2 \cdot 10^5$; $m \leq 5 \cdot 10^5$.

1704 – Network Renovation

A connected network is initially a tree. Add the minimum number of new undirected connections so the network remains connected after the failure of *any one* connection.

In: n ; then $n-1$ tree edges

Out: minimum k , then k added edges; any optimum solution

Constraints: $3 \leq n \leq 10^5$.

1705 – Forbidden Cities

A connected undirected graph is fixed. For each query (a, b, c) , determine whether there is a route from a to b that never visits forbidden city c . **In:** n m q ; graph edges; then queries a b c

Out: YES/NO per query

Constraints: $n, q \leq 10^5$; $m \leq 2 \cdot 10^5$.

1752 – Creating Offices

In a tree, choose as many office vertices as possible so the distance between every pair of chosen vertices is at least d . **In:** n d ; then $n-1$ tree edges

Out: maximum count, then any optimal set of cities

Constraints: $n, d \leq 2 \cdot 10^5$.

1685 – New Flight Routes

Add the minimum number of directed flights to a directed graph so every city becomes reachable from every other city. **In:** n m ; then m directed edges

Out: minimum k , then k added flights; any optimum solution

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

CSES — Counting Problems Reference

All tasks: 1 s, 512 MiB. Unless stated otherwise, answers are modulo $10^9 + 7$.

1075 — Counting Permutations

Count permutations p of $1, \dots, n$ such that no adjacent pair differs by 1, i.e. $|p_i - p_{i+1}| \neq 1$ for every i . **In:** $\mathbf{n}$

Out: number of beautiful permutations mod $10^9 + 7$

Constraints: $1 \leq n \leq 1000$.

1078 — Grid Paths II

In an $n \times n$ grid, move from $(1, 1)$ to (n, n) using only right/down moves while avoiding m trapped cells. Start/end are free and traps are distinct. **In:** $\mathbf{n \ m}$; then m traps $\mathbf{y \ x}$

Out: number of valid paths mod $10^9 + 7$

Constraints: $n \leq 10^6$, $m \leq 1000$, $1 \leq y, x \leq n$.

1080 — Empty String

Given a lowercase string, repeatedly remove any adjacent equal pair. Count the possible sequences of removals that delete the entire string. **In:** lowercase string s

Out: number of ways mod $10^9 + 7$

Constraints: $1 \leq |s| \leq 500$.

2176 — Counting Bishops

Place exactly k bishops on an $n \times n$ chessboard so that no two share a diagonal. Count all placements. **In:** $\mathbf{n \ k}$

Out: number of valid placements mod $10^9 + 7$

Constraints: $1 \leq n \leq 500$, $1 \leq k \leq n^2$.

2228 — Counting Sequences

Count length- n sequences over $\{1, \dots, k\}$ in which every value $1, \dots, k$ appears at least once. **In:** $\mathbf{n \ k}$

Out: number of sequences mod $10^9 + 7$

Constraints: $1 \leq k \leq n \leq 10^6$.

2229 — Permutation Inversions

Count permutations of $1, \dots, n$ having exactly k inversions, where an inversion is a pair $i < j$ with $p_i > p_j$. **In:** $\mathbf{n \ k}$

Out: number of permutations mod $10^9 + 7$

Constraints: $1 \leq n \leq 500$, $0 \leq k \leq n(n-1)/2$.

2415 — Functional Graph Distribution

A functional graph has labeled nodes and outdegree exactly 1 at every node. For every $k = 1, \dots, n$, count such graphs with exactly k connected components. **In:** $\mathbf{n}$

Out: n lines: answer for each $k = 1, \dots, n$ mod $10^9 + 7$

Constraints: $1 \leq n \leq 5000$.

2421 — Counting Reorders

Reorder the multiset of characters of a lowercase string so that adjacent characters are never equal. Count distinct resulting strings. **In:** lowercase string s

Out: number of valid reorderings mod $10^9 + 7$

Constraints: $1 \leq |s| \leq 5000$.

2429 — Grid Completion

Complete an $n \times n$ grid of $\cdot, \mathbf{A}, \mathbf{B}$ so every row and column contains exactly one $\mathbf{A}$ and exactly one $\mathbf{B}$. Existing letters are fixed; initially each row/column has at most one of each. **In:** $\mathbf{n}$; then n grid rows

Out: number of completions mod $10^9 + 7$

Constraints: $2 \leq n \leq 500$.

3157 — Collecting Numbers Distribution

For a permutation of $1, \dots, n$, repeatedly scan left-to-right, collecting the next required values in increasing order until the scan ends. For every k , count permutations requiring exactly k scans. **In:** $\mathbf{n}$

Out: n answers for $k = 1, \dots, n$, mod $10^9 + 7$

Constraints: $1 \leq n \leq 5000$.

3232 — Tournament Graph Distribution

A tournament has exactly one directed edge between each pair of labeled nodes. For every $k = 1, \dots, n$, count tournaments with exactly k strongly connected components. **In:** $\mathbf{n}$

Out: n lines: answer for each $k = 1, \dots, n$ mod $10^9 + 7$

Constraints: $1 \leq n \leq 500$.

3400 — Raab Game II

Each player owns cards $1, \dots, n$ and plays every card exactly once over n turns. Higher card scores 1 point, ties score neither; count ordered pairs of play orders giving final scores (a, b) . **In:** $\mathbf{t}$; then t tests $\mathbf{n \ a \ b}$

Out: answer for each test mod $10^9 + 7$

Constraints: $t \leq 1000$, $n \leq 5000$, $0 \leq a, b \leq n$.

3413 — Filled Subgrid Count I

Grid entries are the first k uppercase letters. For each letter, count contiguous axis-aligned *square* subgrids whose every cell equals that letter. **In:** $\mathbf{n \ k}$; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3414 — Filled Subgrid Count II

For each of the first k uppercase letters, count contiguous axis-aligned *rectangular* subgrids whose every cell equals that letter. **In:** $\mathbf{n \ k}$; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3415 — All Letter Subgrid Count I

Count contiguous axis-aligned *square* subgrids containing every one of the first k uppercase letters at least once. **In:** $\mathbf{n \ k}$; then n grid rows

Out: exact number of qualifying squares

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3416 — All Letter Subgrid Count II

Count contiguous axis-aligned *rectangular* subgrids containing every one of the first k uppercase letters at least once. **In:** $\mathbf{n \ k}$; then n grid rows

Out: exact number of qualifying rectangles

Constraints: $n \leq 500$, $k \leq 26$. Use 64-bit.

3417 — Border Subgrid Count I

For each letter, count contiguous axis-aligned *square* subgrids whose entire border consists of that letter; interior cells may be arbitrary. **In:** $\mathbf{n \ k}$; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 3000$, $k \leq 26$. Use 64-bit.

3418 — Border Subgrid Count II

For each letter, count contiguous axis-aligned *rectangular* subgrids whose entire border consists of that letter; interior cells may be arbitrary. **In:** $\mathbf{n \ k}$; then n grid rows

Out: k exact counts, one per letter

Constraints: $n \leq 500$, $k \leq 26$. Use 64-bit.

CSES — Additional Problems II Reference

All tasks: 1 s, 512 MiB. Modulo $10^9 + 7$ only where explicitly stated.

1148 — Maximum Building II

Given an $n \times m$ forest grid of empty cells `.` and trees `*`, count placements of an all-empty axis-aligned rectangle for *every exact size* $h \times w$.

In: `n` `m`; then n grid rows

Out: n rows; row h contains the counts for widths $w = 1, \dots, m$

Constraints: $1 \leq n, m \leq 1000$.

1157 — Mex Grid Queries

An infinite positive-index grid is filled so cell (y, x) is the smallest nonnegative integer absent both to its left in row y and above it in column x . Find the value at one queried cell. **In:** `y` `x`

Out: value at (y, x)

Constraints: $1 \leq y, x \leq 10^9$.

1159 — Book Shop II

Book i costs h_i , gives s_i pages, and has at most k_i copies available. With total cost at most x , maximize purchased pages. **In:** `n` `x`; prices h ; pages s ; copy limits k

Out: maximum total pages

Constraints: $n \leq 100$, $x \leq 10^5$, $h_i, s_i, k_i \leq 1000$.

1161 — Stick Divisions

Start with one stick of length x . A move splits one current stick into two and costs the original stick length; end with the multiset of target lengths $d_1, \dots, d_n$. Minimize total cost. **In:** `x` `n`; $d_1 \dots d_n$

Out: minimum total split cost

Constraints: $x \leq 10^9$, $n \leq 2 \cdot 10^5$, $\sum d_i = x$. Use 64-bit.

1189 — Food Division

Children form a cycle. Child i initially has a_i food and must end with b_i ; one step transfers one unit to either neighbor. The total food is already correct. Minimize transfers. **In:** `n`; $a_1 \dots a_n$; $b_1 \dots b_n$

Out: minimum number of unit transfers

Constraints: $n \leq 2 \cdot 10^5$, $a_i, b_i \leq 10^6$. Use 64-bit.

1665 — Coding Company

Partition the n coders into teams. A team's penalty is $\max t - \min t$; count divisions whose sum of team penalties is at most x . **In:** `n` `x`; skills $t_1 \dots t_n$

Out: number of valid divisions mod $10^9 + 7$

Constraints: $n \leq 100$, $x \leq 5000$, $0 \leq t_i \leq 100$.

1698 — Swap Round Sorting

Sort a permutation in the minimum number of *swap rounds*; within one round, choose any number of distinct pairs and swap each chosen pair. Also construct the swaps in every round. **In:** `n`; permutation p

Out: minimum round count; for each round, its swaps as index pairs

Constraints: $n \leq 2 \cdot 10^5$.

1706 — School Excursion

Each given pair of children insists on visiting the same attraction, so every connected component must stay together. For each $i = 1, \dots, n$, decide whether exactly i children can be assigned to Korkeasaari. **In:** `n` `m`; then m same-attraction pairs

Out: length- n bit string; bit i is 1 iff size i is possible

Constraints: $n, m \leq 10^5$.

1709 — Coin Grid

An $n \times n$ binary coin grid can be cleared by a move that removes all remaining coins in one chosen row or column. Find the minimum number of moves and output any optimal set of rows/columns. **In:** `n`; then n rows over `.`, `o`

Out: minimum k , then k moves 1 `r` (row) or 2 `c` (column)

Constraints: $1 \leq n \leq 100$.

2075 — Reversal Sorting

Sort a permutation by reversing contiguous subarrays. Any construction using at most n reversals is accepted; the number of reversals need not be minimum. **In:** `n`; permutation p

Out: $k \leq n$, then reversal intervals $[a, b]$

Constraints: $n \leq 2 \cdot 10^5$.

2115 — Bit Substrings

For each $k = 0, 1, \dots, n$, count the *nonempty* substrings of a bit string that contain exactly k ones. **In:** binary string s

Out: $n + 1$ counts for $k = 0, \dots, n$

Constraints: $|s| \leq 2 \cdot 10^5$. Counts need 64-bit.

2131 — Grid Puzzle II

Choose grid cells so row i contains exactly a_i chosen cells and column j exactly b_j . Cell (i, j) yields c_{ij} coins; maximize the collected sum and construct the chosen cells, or detect infeasibility. **In:** `n`; row quotas a ; column quotas b ; then coin matrix c

Out: maximum sum and an `X/. grid`, or `-1`

Constraints: $n \leq 50$, $a_i, b_j \leq n$, $c_{ij} \leq 1000$; $\sum a_i = \sum b_j$.

2132 — Increasing Array II

Change each array value by unit increments/decrements until the sequence is nondecreasing. Minimize the total absolute change. **In:** `n`; $x_1 \dots x_n$

Out: minimum number of unit changes

Constraints: $n \leq 2 \cdot 10^5$, $x_i \leq 10^9$. Use 64-bit.

2174 — Removing Digits II

Starting from n , repeatedly subtract any one decimal digit currently appearing in the number. Find the minimum steps to reach 0. **In:** `n`

Out: minimum number of steps

Constraints: $1 \leq n \leq 10^{18}$. Use 64-bit.

2180 — Coin Arrangement

A $2 \times n$ grid contains exactly $2n$ coins total, possibly multiple per cell. Move individual coins one orthogonal step per operation until every cell has exactly one; minimize total moves. **In:** `n`; then the two rows of coin counts

Out: minimum number of moves

Constraints: $1 \leq n \leq 10^5$. Use 64-bit.

2402 — Two Stacks Sorting

Process a permutation from left to right using two stacks: move the first remaining input value onto one stack, or pop a stack value to the output. The final output must be $1, 2, \dots, n$. **In:** $\mathbf{n}$; input permutation
Out: for each input number, a chosen stack 1 or 2; or IMPOSSIBLE
Constraints: $n \leq 2 \cdot 10^5$.

2426 — Programmers and Artists

Each applicant has programming skill x and artistic skill y . Hire exactly a programmers and b artists, with no applicant used twice, maximizing total skill. **In:** $\mathbf{a} \ \mathbf{b} \ \mathbf{n}$; then n pairs $\mathbf{x} \ \mathbf{y}$
Out: maximum total skill
Constraints: $n \leq 2 \cdot 10^5$, $a + b \leq n$, $x, y \leq 10^9$. Use 64-bit.

2430 — Binary Subsequences

Construct a shortest bit string having exactly n distinct *nonempty* subsequences (identical resulting strings count once). Any minimum-length construction is accepted. **In:** $\mathbf{n}$
Out: any minimum-length valid bit string
Constraints: $1 \leq n \leq 10^6$.

2432 — Grid Puzzle I

Choose cells of an $n \times n$ grid so row i has exactly a_i chosen cells and column j exactly b_j . Construct any feasible $\mathbf{X}/.$ grid. **In:** $\mathbf{n}$; row quotas a ; column quotas b
Out: a valid grid, or $\mathbf{-1}$ if impossible
Constraints: $n \leq 50$, $0 \leq a_i, b_j \leq n$.

3108 — K Subset Sums I

Consider all 2^n subsets, including the empty subset. Output the k smallest subset sums in nondecreasing order; equal sums from different subsets appear with multiplicity. **In:** $\mathbf{n} \ \mathbf{k}$; $x_1 \dots x_n$
Out: k smallest subset sums
Constraints: $n \leq 2 \cdot 10^5$, $k \leq 2 \cdot 10^5$, $|x_i| \leq 10^9$. Use 64-bit sums.

3109 — K Subset Sums II

Consider all subsets containing exactly m array elements. Output the k smallest sums in nondecreasing order, retaining multiplicity between different subsets. **In:** $\mathbf{n} \ \mathbf{m} \ \mathbf{k}$; $x_1 \dots x_n$
Out: k smallest size- m subset sums
Constraints: $m < n \leq 2 \cdot 10^5$, $k \leq 2 \cdot 10^5$, $|x_i| \leq 10^9$. Use 64-bit sums.

3159 — Replace with Difference

Repeatedly choose two *current numbers* a, b , delete both, and insert $|a - b|$. Construct $n - 1$ operations so the sole remaining value is 0, or report impossibility. **In:** $\mathbf{n}$; $x_1 \dots x_n$
Out: chosen value pairs a, b for every operation, or $\mathbf{-1}$
Constraints: $2 \leq n \leq 1000$, $1 \leq x_i \leq 1000$.

3161 — GCD Subsets

For every $k = 1, \dots, n$, count nonempty subsets of array positions whose selected values have greatest common divisor exactly k . **In:** $\mathbf{n}$; $x_1 \dots x_n$
Out: n answers for $\gcd 1, \dots, n$, modulo $10^9 + 7$
Constraints: $n \leq 2 \cdot 10^5$, $1 \leq x_i \leq n$.

3215 — Bouncing Ball Steps

A ball starts at (1,1) in an $n \times m$ grid moving diagonally down-right, one cell per step, reflecting whenever it hits a border. For a huge k , find its position and total number of direction changes after k steps. **In:** $\mathbf{t}$; then tests $\mathbf{n} \ \mathbf{m} \ \mathbf{k}$
Out: for each test: row, column, direction-change count
Constraints: $t \leq 1000$, $n, m \leq 10^9$, $k \leq 10^{18}$. Use 64-bit.

3216 — Bouncing Ball Cycle

For the same reflecting diagonal motion, determine when the ball first reaches (1,1) again and how many distinct grid cells were visited during the cycle. **In:** $\mathbf{t}$; then tests $\mathbf{n} \ \mathbf{m}$
Out: for each test: return time and number of distinct visited cells
Constraints: $t \leq 1000$, $2 \leq n, m \leq 10^9$. Use 64-bit.

3218 — Knight Moves Queries

On the infinite chessboard with positive row/column indices, answer the minimum knight moves from each queried (x, y) to the top-left corner (1,1). **In:** $\mathbf{q}$; then q positions $\mathbf{x} \ \mathbf{y}$
Out: minimum knight distance for each query
Constraints: $q \leq 10^5$, $x, y \leq 10^9$.

3312 — Grid Coloring II

Every cell initially contains A, B, or C. Change *every* cell to a different letter so that no orthogonally adjacent final cells are equal. Construct any solution. **In:** $\mathbf{n} \ \mathbf{m}$; then n grid rows
Out: a valid recolored grid, or IMPOSSIBLE
Constraints: $n, m \leq 500$.

3401 — Stick Difference

Make exactly k cuts among integer-length sticks, producing $n + k$ positive-integer pieces. For every $k = 1, \dots, m$, minimize $\max(\text{piece}) - \min(\text{piece})$. **In:** $\mathbf{n} \ \mathbf{m}$; lengths $a_1 \dots a_n$
Out: m minima for exactly $k = 1, \dots, m$ cuts
Constraints: $n \leq 10^5$, $m \leq 2 \cdot 10^5$, $a_i \leq 10^9$; all m cuts are feasible.

3402 — Minimum Cost Pairs

For each $k = 1, \dots, \lfloor n/2 \rfloor$, choose k disjoint pairs of array elements. Pair (a, b) costs $|a - b|$; minimize the total cost. **In:** $\mathbf{n}$; $x_1 \dots x_n$
Out: $\lfloor n/2 \rfloor$ minimum costs
Constraints: $n \leq 2 \cdot 10^5$, $x_i \leq 10^9$. Use 64-bit.

3425 — Same Sum Subsets

Choose two disjoint nonempty subsets of the given positive elements whose sums are equal. Output any valid pair of subsets. **In:** $\mathbf{n}$; $x_1 \dots x_n$
Out: for each subset, its size and elements; or IMPOSSIBLE
Constraints: $3 \leq n \leq 40$, $\sum x_i \leq 2^n - 2$.